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# Voting on sanctioning institutions in open and closed communities: experimental evidence

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## Abstract

We experimentally analyze the effect of endogenous group formation on the type of sanctioning institutions emerging in a society. We allocate subjects to one of two groups. Subjects play a repeated public goods game and vote on the sanctioning system (formal or informal) to be implemented in their group. We compare this environment to one in which subjects are allowed to (i) vote on the sanctioning system and (ii) move between groups. We find that the possibility of moving between groups leads to a larger proportion of subjects voting for formal sanctions. This result is mainly driven by subjects in groups with relatively high initial levels of contribution to the public good, who are more likely to vote for informal sanctions when groups are closed than when they are open.

## 1 Introduction

The presence and functioning of sanctioning institutions is key to overcoming free-riding problems (Ostrom et al. 1992; Fehr and Gächter 2000). Sanctioning institutions can take different forms, and most societies are involved in a dynamic process

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of institutional adaptation. A sizable literature has recently recognized the relevance of the endogenous choice of sanctioning institutions (e.g. Ertan et al. 2009; Sutter et al. 2010; Andreoni and Gee 2012). In particular, increasing attention has been devoted to the choice between decentralized, peer-to-peer sanctions, and more centralized and formalized forms of punishment. Several studies find considerable support for peer-to-peer sanctions, even when centralized sanctions are theoretically expected to prevail (Markussen et al. 2014; Zhang et al. 2014; Kamei et al. 2015; Nicklisch et al. 2016). This paper asks whether this result continues to hold when societies are open to entry and exit. In other words, we investigate whether open societies implement different sanctioning institutions than closed ones.

The topic is important in a number of contexts where mobility across group boundaries is changing. In developing countries, population growth, improvements in infrastructure and changing economic opportunities lead to increased internal migration, for example from rural to urban areas (Harris and Todaro 1970; Deshingkar and Anderson 2004). In large organizations such as companies, the increasing use of temporary employment (cf. the "gig economy") means that the composition of work teams changes at more rapid pace than before (De Stefano 2016). And at the international level, wars, political oppression and gaps in economic opportunities lead many to migrate across borders. The experiment presented here contributes to our understanding of how increased mobility changes preferences between informal, horizontal institutions, and more centralized, formal arrangements, such as the rules and regulations that a state or the management of a large organization may implement.[1]

There are two main approaches to studying the choice between sanctioning institutions in social dilemmas. In one strand of the literature, groups of fixed size and composition vote by ballot about which type of institution to use (e.g. Sutter et al. 2010; Andreoni and Gee 2012; Markussen et al. 2014). Another cluster of experiments allows participants to "vote with their feet" by migrating to the institutional environment they prefer (e.g. Gürerk et al. 2006, 2014; Nicklisch et al. 2016; Zhang et al. 2014).[2] We combine these two approaches. In the baseline treatment, subjects are randomly allocated to two groups of 5 people to play a Public Goods Game (PGG). Subjects interact for 30 periods and group size and composition remain fixed for the entire duration of the experiment. Subjects vote every five periods on the sanctioning institutions—either formal or informal—to be implemented in their group. With informal sanctions, participants receive information about the contribution decisions of their peers and are given the opportunity to reduce the earnings of other group members, at a cost

[1] A relevant historical example is merchant guilds in Europe (Ogilvie 2011; Dessi and Piccolo 2016). We may view the guilds as a type of (relatively) informal, horizontal institution. Around 1800, as trade and mobility were expanding in the wake of the industrial revolution, many guilds were broken up. To a large extent, they were replaced with government regulation such as patent systems. Hence, this is an example of a move toward more formal, centralized institutions, which coincided with the emergence of a more dynamic, open society.

[2] As discussed below, exceptions include Robbett (2014) and Cobo-Reyes et al. (2019). For a survey of the literature, see Dannenberg and Gallier (2020).

to themselves (cf. Fehr and Gächter 2000). With formal sanctions, free riders are automatically punished. The formal sanctioning system comes with a fixed cost, which must be paid by all group members. We compare the baseline with our main treatment, in which subjects not only vote about institutional choice, but can also (freely) move between groups (cf. Tiebout 1956). In other words, we compare an environment where both "voice" (voting by ballot) and "exit" (voting with feet) strategies are available—i.e., "open societies"—to one where only the "voice" option exists—i.e., "closed societies"—(cf. Hirschman 1970).

A priori, it is far from clear whether one should expect individuals to prefer more or less centralized forms of sanctioning institutions when groups' size and composition can vary. From a theoretical point of view, Fehr and Schmidt (1999) show that if people have sufficiently strong social preferences (in particular, aversion to inequality), then multiple equilibria exist in public goods games with decentralized, informal sanctions (IS), some of them with high levels of cooperation and some with low levels. Reactions to opening borders may differ markedly, depending on whether groups have previously converged to cooperative or non-cooperative equilibria. On the one hand, if a group has established a cooperative equilibrium, allowing inward migration is a risk factor that may potentially lead the equilibrium to break down. Hence, groups that have established high levels of contributions under informal sanctions should vote for continued use of IS when their size and composition is fixed, but may prefer centralized, formal sanctions (FS) when migration is allowed, because the influx of new members potentially leads to the breakdown of cooperative equilibria. On the other hand, if cooperative norms have not been established, inward and outward migration could either provide an opportunity to move to a better equilibrium or it could make cooperation even harder to achieve.

Informed by a formal theoretical framework, we hypothesize that IS will lead to heterogeneous outcomes across groups, and that the contributions to the public good under IS will have a strong impact on how the opportunity to move between groups affects voting patterns. To test this hypothesis, we fix groups during the first five periods of the experiment and impose IS (rather than FS) in all of them. After this initial phase, voting between IS and FS is introduced and the opportunity to move between groups is made available in one treatment.

We find that endogenous group formation dramatically affects institutional choice, but in different ways depending on groups' initial conditions. In particular, in our baseline treatment, subjects in groups with high average contributions to the public good in the initial phase of the experiment vote in favor of Informal Sanctions more than 80% of the time; this proportion decreases to less than 55% when subjects can move between groups. When subjects start out in groups with low average contributions to the public good in the initial phase of the experiment, the opposite results hold. In this case, 33% of the subjects vote for Informal Sanctions when groups are fixed, whereas more than 45% of the subjects choose Informal Sanctions when they can move between groups. Results also show that, when given the opportunity, on average 20% of the subjects move between groups in each period. Subjects' group choice is mainly determined by the difference in contribution between the subject and other members of her group, and by the punishment she receives. We also find that although contributions are on average larger when Formal Sanctions are in place, this difference does not affect migration behavior and subjects do not tend to move systematically towards societies implementing Formal Sanctions.

Only a few other experimental studies on social dilemmas combine “voting by ballot” and “voting with feet” in the same experimental set-up. Robbett (2014) implements a non-linear public goods game and studies the choice of taxes, rather than sanctioning systems. She compares a treatment where only voting with feet is possible to one where both voting with feet and by ballots is allowed. In contrast, our treatments differ in the availability of the voting with feet option. This is how we distinguish “open” from “closed” societies. We focus on the choice of institutions, whereas Robbett is mainly interested in efficiency.[3] Cobo-Reyes et al. (2019) study the choice between formal sanctions and no sanctions at all in a public goods game but, like Robbett (2014), vary only the availability of the voting by ballot option, while voting with feet is always allowed. Hence, none of these papers compare voting about institutions in open and closed societies. Doing so is the main focus of this paper.

The paper proceeds as follows. Section 2 describes the experimental design, introduces a formal model, and presents the hypotheses to be tested in the empirical analysis. Section 3 reports and discusses our findings. Section 4 concludes. All proofs are relegated to the Appendix.

## 2 Experimental procedures and hypotheses

### 2.1 Experimental design

We conducted laboratory experiments at the University of Exeter between October 2017 and February 2018. Subjects were mainly students of economics, business administration, and engineering, but other disciplines were also represented.[4] We ran two different treatments with a total of 20 sessions, yielding 10 independent observations (sessions) per treatment. Each session lasted 30 rounds and included 10 participating subjects, with 6000 individual-per-period observations across all sessions. Subjects were paid for three randomly selected periods, and the average individual earnings were £14. All instructions can be found in Appendix A.

The experimental design consists of two treatments: the No-Moving Treatment (NMT) and the Moving Treatment (MT). Treatments differ depending on whether the groups are endogenously formed.

[3] Salmon and Weber (2017) also study an environment in which individuals can move between groups. Their focus is on the effectiveness of alternative rules designed to select the individuals who are allowed to enter a given group. We instead focus on how individuals’ institutional choice varies depending on whether migration is restricted or not. Page et al. (2005) show that individuals are significantly more likely to cooperate in a public goods game when allowed to have a say over with whom they interact. The sorting process in their paper is solely based on previous contributions and, unlike us, the authors do not analyze the role played by different institutional settings.

[4] Ethics approval for experimentation with human subjects was provided by the University of Exeter.

**No Moving Treatment (NMT)** In the first period, subjects are randomly assigned to one of two groups—A and B—of 5 people each. Each period is composed of two (sequential) stages: a *contribution stage* and a *punishment stage*. In the *contribution stage*, each subject decides how to allocate the 50 tokens she is endowed with between a group account and a private account.

There are two institutional settings (rule-sets) affecting subjects' payoffs.

(a) Under **IS**, informal sanctioning institutions are in place. In the *punishment stage*, subjects observe the contributions to the group account made by their fellow group members. They then have the opportunity to reduce the earnings of the other members of their group. Specifically, any subject $i$ can impose a sanction of 3 units on any other group member $j$ at a cost of 1 unit to herself. A subject can spend at most 10 tokens on punishing any other. The monetary payoff of subject $i$ under IS is:

$$\pi_i = (50 - C_i) + \frac{1.6}{n^g}\left(\sum_{j=1}^{n^h} C_j\right) - \sum_{j=1}^{n^g} R_{i,j} - 3\sum_{j=1}^{n^g} R_{j,i}, \tag{1}$$

for $i, j = 1, ..., n^g$ and $g = A, B$, where $n^g$ denotes the total number of subjects located in group $g$, and $C_i \in [0, 50]$ denotes $i$'s contribution to the group account. We denote by $R_{i,j}$ ($R_{j,i}$) the number of tokens that subject $i$ ($j$) uses to punish subject $j$ ($i$).[5] Finally, the factor multiplying the contributions to the group account is equal to 1.6, implying a marginal per-capita return from the group-$h$ account equal to $\frac{1.6}{n^g}$.

(b) Under **FS**, formal sanctions are in place. Each subject in the group pays a fixed fee of 5 tokens per period. In addition, in the *punishment stage*, each subject pays a fine equal to 80% of the amount of tokens allocated to the private account in a given period. The fixed fee and the fine (if applicable) are deducted from subjects' monetary payoff. Therefore, under FS, subject $i$'s per-period monetary payoff is:

$$\pi_i = (50 - C_i)(1 - 0.8) + \frac{1.6}{n^g}\left(\sum_{j=1}^{n^h} C_j\right) - 5, \tag{2}$$

for $i, j = 1, ..., n^g$ and $g = A, B$.[6]

At the end of the *punishment stage*, each subject learns her payoff. In IS, subjects also learn the amount of punishment they receive, but not who punished them or how much punishment others receive in total. In FS, subjects are informed of the amount of the fine they pay. In both institutional settings, participants also receive information about: (i) the average contribution to the group account in their current

[5] When $n^g = 1$, the lone subject in a group simply receives her endowment of 50 tokens.

[6] Note that in the experimental instructions (Appendix A) we refer to the IS (respectively, FS) rule-sets as Rule Set 1 (respectively, Rule Set 2).

group; (ii) the average payoffs in their current group; (iii) the average payoffs in the other group; (iv) the rule-set implemented in their current group; and (v) the rule-set implemented in the other group.

In the first 5 periods, both Group A and B use IS. Starting from period 6, subjects vote every 5 periods on the rule-set to be implemented in their group. Subjects therefore vote 5 times in total, with the *voting stage* taking place at the beginning of the period. The rule-set is implemented immediately after voting and applies until the next voting round. Hence, from period 6 onward, subjects first decide on the rule-set (if the period includes a *voting stage*), then contribute to the group account (*contribution stage*), and finally participate in the *punishment stage*.

**Moving Treatment** (**MT**) This treatment differs from NMT in one aspect only. In NMT, subjects cannot move between groups: there are two groups of 5 members that are fixed for the entire duration of the experiment. In MT, by contrast, Group A and B are fixed in the first 5 periods, but starting from period 6 moving between groups is allowed at the end of each period. Participants are made aware of this before the start of the experiment, so they know in the first five periods that they will be able to move from the 6th period on. Subjects enter the *moving stage* after the *punishment stage*, and simultaneously decide whether to move from their current group. Hence, from period 6 onwards, subjects first decide on the rule-set to be implement in their current group (if the period includes a *voting stage*), then contribute to the group account (*contribution stage*), then participate in the *punishment stage*, and finally decide whether to move.[7]

As explained in the Introduction, the decision to keep groups fixed and let all groups use IS during the first 5 periods of the experiment was taken in order to allow groups to develop distinct, cooperative norms in the IS environment. This is necessary in order to test our main hypothesis that reactions to opening up the possibility of migration depend on pre-existing norms of cooperation.

We now introduce a simple formal model that allows us to motivate our hypotheses.

### 2.2 Theoretical setting and hypotheses

We construct a simple theoretical framework that captures the main features of our experimental design.[8]

All proofs are relegated to Appendix B.

We consider two groups composed of $n^g$ individuals, with $g \in \{A, B\}$ denoting the group index, and where $n = n^A + n^B$ represents the total number of individuals in our economy. Individuals play a public goods game (henceforth, *PGG*). Let $E$ denote individual $i$'s endowment, for $i = 1, ..., n$. Each individual chooses a contribution

[7] In MT, subjects are informed about their current group size at the beginning of each period.

[8] For simplicity, we consider a one-period game in our theoretical setting. The finitely repeated multi-period game in our experimental design accounts for the possibility that convergence to an equilibrium may only occur after some iterations of the game.

$C_i \in [0, E]$ to the public good, where $E - C_i$ denotes $i$'s consumption of a private good. Individual $i$'s monetary payoff is:

$$\pi_i = E - C_i + \left[ \frac{a}{n^g} \sum_{j=1}^{n^g} C_j \right] \mathbb{1}_{n^g > 1}, \tag{3}$$

with $a > 1$, $\frac{a}{n^g} < 1$ for all values $n^g > 1$, and where $\mathbb{1}_{n^g > 1}$ is an indicator function that takes value 1 if $n^g > 1$ and 0 otherwise, for $i = 1, ..., n^g$ and $g \in \{A, B\}$. While contributing $C_i = E$ is socially efficient ($a > 1$), each individual $i$ has an incentive to free-ride—i.e., the marginal per capita return (MPCR) $\frac{a}{n^g}$ is lower than one, for all values $n^g > 1$.

Individuals can interact under two possible institutional settings.

- *Informal Sanctions (IS)* Under IS, individual $i$ who is located in group $g$ can impose a sanction $c$ on any other individual $j$ located in the same group at a cost of one unit to herself. Let us denote by $R_{i,j}$ $i$'s cost of punishing $j$; similarly, we define $cR_{j,i}$ as the cost to $i$ of being punished by $j$, for $i, j \in 1, ..., n^g$, for $g \in \{A, B\}$. Therefore, under IS, $i$'s monetary payoff is:

$$\pi_i = E - C_i + \left[ \frac{a}{n^g} \sum_{j=1}^{n^g} C_j \right] \mathbb{1}_{n^g > 1} - \sum_{j=1}^{n^g} R_{i,j} - c \sum_{j=1}^{n^g} R_{j,i}, \tag{4}$$

  for $i = 1, ..., n^g$ and $g \in \{A, B\}$.
- *Formal Sanctions (FS)* Under FS, individual $i$ pays a fixed fee equal to $F$. In addition, individual $i$ incurs a sanction equal to a share $s \in (0, 1)$ of the amount she allocates to the consumption of the private good. Therefore, under FS, $i$'s monetary payoff is:

$$\pi_i = \left[E - C_i\right](1 - s) + \left[ \frac{a}{n^g} \sum_{j=1}^{n^g} C_j \right] \mathbb{1}_{n^g > 1} - F, \tag{5}$$

  for $i = 1, ..., n^g$ and $g \in \{A, B\}$. In what follows, we assume that formal sanctions are *deterrent*, that is, $s \geq 1 - \frac{a}{n^g}$, for $g \in \{A, B\}$ and $n^g > 1$. Under *deterrent* formal sanctions, rational and self-interested individuals find it profitable to contribute all of their endowments to the public good. Moreover, as in Markussen et al. (2014) and in line with our experimental design, we assume $F < aE - E$, where we define the right-hand-side of the inequality as the "cooperation premium".

In our experiment, treatments vary depending on whether individuals are allowed to (freely) move between groups. We consider these two cases separately.

### 2.2.1 No moving

Consider the case in which individuals are not allowed to move between groups.

In line with our experimental design, we assume $n^A = n^B$, where $n^A$ is an odd number. Individuals interact for one period. At the beginning of the period, individuals can vote over the type of sanctioning institutions—either IS or FS—to be implemented in their group. We denote by $v_i \in \{0,1\}$ individual $i$'s vote, where $v_i = 1$ (respectively, $v_i = 0$) denotes individual $i$'s vote in favor of implementing IS (respectively, FS) in her group $g$, for $i = 1, ..., n^g$ and $g \in \{A, B\}$. Sanctioning institutions are selected by majority voting. Once sanctioning institutions are in place, individuals choose their contributions to the public good. A punishment stage follows, with the punishment varying depending on the type of sanctioning institutions in place. More formally, the timing of the game is:

1. *Voting* Individuals simultaneously choose $v_i \in \{0,1\}$, for $i = 1, ..., n^g$ and $g \in \{A, B\}$;
2. *Contribution* In each group, sanctioning institutions that obtain the majority of votes by group members are implemented. Individuals observe the institutions in place and simultaneously choose $C_i \in [0, E]$, for $i = 1, ..., n^g$ and $g \in \{A, B\}$;
3. *Punishment* If FS are in place, punishment is automatically applied. If IS are in place, individuals observe all contributions to the public good in their group and simultaneously choose punishments $R_{i,j}$, for $i, j = 1, ..., n^g$, $i \neq j$ and $g \in \{A, B\}$;
4. *Payoff*: payoffs realize.

We state our predictions under (*i*) standard (selfish) preferences and (*ii*) social preferences as in Fehr and Schmidt (1999).[9]

Our equilibrium concept is sub-game perfection.

*Selfish preferences* Suppose it is common knowledge that all individuals are rational and self-interested. The following result holds.

**Proposition 1** *Suppose individuals cannot move between groups. Then, under selfish preferences, individuals implement formal sanctions in their group and contribute* $C_i^* = E$, *for* $i = 1, ..., n^g$ *and* $g \in \{A, B\}$.

Under selfish preferences, because of free-riding at the punishment stage, informal sanctions are ineffective at supporting contributions to the public good. Therefore, individuals implement formal sanctions to prevent free-riding at the contribution stage.

*Social Preferences* Suppose that some individuals are averse to inequality. Following Fehr and Schmidt (1999), individual $i$'s utility is given by:

$$U_i = \pi_i - \alpha_i \frac{1}{n^g - 1} \sum_{j \neq i} \max\left[\pi_j - \pi_i, 0\right] - \beta_i \frac{1}{n^g - 1} \sum_{j \neq i} \max\left[\pi_i - \pi_j, 0\right], \quad (6)$$

[9] This analysis is not intended as a validation of alternative theories of social preferences. We have therefore chosen to adopt the framework proposed by Fehr and Schmidt (1999) because of its parsimony.

for $i, j = 1, \ldots, n^g$, $i \neq j$ and $g \in \{A, B\}$, with $\alpha_i \geq \beta_i$ and $\beta_i \in [0, 1]$. The parameter $\alpha_i$ (respectively, $\beta_i$) denotes individual $i$'s aversion to disadvantageous (respectively, advantageous) inequality.[10] From Fehr and Schmidt (1999) and Markussen et al. (2014), the following two results hold.

**Lemma 1** *Suppose IS are in place in a group g, for* $g \in \{A, B\}$. *Suppose also there is a group of* $n'$ *"conditionally cooperative enforcers",* $1 \leq n' \leq n^g$, *with preferences that obey* $\frac{a}{n^g} + \beta_i \geq 1$ *and*

$$\frac{1}{c} < \frac{\alpha_i}{(n-1)\left(1+\alpha_i\right) - \left(n'-1\right)\left(\alpha_i+\beta_i\right)}, \tag{7}$$

*for* $i \in \{1, \ldots, n'\}$, *whereas* $\alpha_i = \beta_i = 0$, *for* $i \in \{n'+1, \ldots n^g\}$ *and* $g \in \{A, B\}$. *Then, there exist equilibria in which* $C_i = C \in [0, E]$ *and no punishment occurs.*

**Lemma 2** *Suppose FS are in place in a group g, for* $g \in \{A, B\}$. *Then, there exists an equilibrium in which* $C_i = E$, *for* $i = 1, \ldots, n^g$ *and* $g \in \{A, B\}$.

In the presence of social preferences, in contrast to what standard theory would predict, there exist equilibria in which individuals make positive contributions to the public good under IS (Lemma 1). Lemma 2 confirms that full contributions to the public good occur under FS.

Let us define $\hat{C} \in [0, E]$ as the equilibrium level of contributions in the sub-game starting under IS (see Lemma 1). From (6), there exists a threshold $\tilde{C} = E - \frac{F}{a-1}$ such that individual $i$ prefers IS to FS (respectively, FS to IS) if $\hat{C} \geq \tilde{C}$ (respectively, $\hat{C} < \tilde{C}$). Further, we define as "high contribution group" (respectively, "low contribution group") a group whose members can (respectively, cannot) support an equilibrium such that $\hat{C} \in \left[\tilde{C}, E\right]$ under IS.[11] We can now state the following result.

**Proposition 2** *Suppose individuals cannot move between groups. Then, under social preferences à la (Fehr and Schmidt 1999), individuals in "high contribution groups" (respectively, "low contribution groups") find it optimal to vote in favor of informal sanctions (respectively, formal sanctions) in their group. Individuals in "high contribution groups" (respectively, "low contribution groups") contribute* $C_i^* = C^* \in \left[\tilde{C}, E\right]$ *(respectively,* $C_i^* = C^* = E$*) to the public good, for* $i = 1, \ldots, n^g$ *and* $g \in \{A, B\}$.

In line with Markussen et al. (2014), because social preferences make cooperation possible under IS, groups that manage to establish sufficiently cooperative

[10] For the sake of simplicity, we assume that parameters $\alpha_i$ and $\beta_i$ are common knowledge, for $i = 1, \ldots, n$.

[11] As discussed below, we resort to alternative operationalizations of the notions of "high" and "low" contributions in our empirical analysis. These operationalizations are very similar in spirit—though not necessarily identical—to the concepts developed in our theoretical framework.

norms prefer this decentralized institution over more centralized and costly forms of sanctioning.

### 2.2.2 Moving

Consider the case in which individuals can freely move between groups. We define $\tilde{n}^g$ as the total number of individuals *initially* located in group $g$ (i.e., before *moving* occurs), for $g \in \{A, B\}$. The variable $n^g$ denotes the total number of individuals located in group $g$ after the *moving stage* occurs, for $g \in \{A, B\}$. We denote by $m_i \in \{0, 1\}$ individual $i$'s decision to move, where $m_i = 1$ (respectively, $m_i = 0$) denotes a movement (respectively, no movement) between groups, for $i = 1, ..., \tilde{n}^g$ and $g \in \{A, B\}$. The timing of the game is:[12]

1. *Voting* Individuals simultaneously choose $v_i \in \{0, 1\}$, for $i = 1, ..., \tilde{n}^g$ and $g \in \{A, B\}$;
2. *Moving* Individuals simultaneously choose $m_i \in \{0, 1\}$, for $i = 1, ..., \tilde{n}^g$ and $g \in \{A, B\}$;
3. *Contribution* In each group, sanctioning institutions that obtain the majority of votes by group members are implemented. Individuals observe the institutions in place and simultaneously choose $C_i \in [0, E]$, for $i = 1, ..., n^g$ and $g \in \{A, B\}$;
4. *Punishment* If FS are in place, punishment is automatically applied. If IS are in place, individuals observe all contributions to the public good in their group and simultaneously choose punishments $R_{i,j}$, for $i, j = 1, ..., n^g$, $i \neq j$ and $g \in \{A, B\}$;
5. *Payoff*: payoffs realize.

*Selfish Preferences* Suppose that all individuals are rational and self-interested. The following result holds.

**Proposition 3** *Suppose individuals can move between groups. Then, under selfish preferences, individuals implement formal sanctions in at least one group. Also, all individuals are located in groups that implement formal sanctions and contribute* $C_i^* = E$, *for* $i = 1, ..., n^g$ *and* $g \in \{A, B\}$.

Overall, comparing Propositions 1 and 3, the standard theory predicts that all individuals play under FS, independently of whether they can move between groups.

*Social Preferences* Suppose individuals' utility is given by (6). According to Lemma 1, multiple equilibria exist in the sub-game in which individuals are located in a group implementing IS. As discussed in Proposition 2, if the level of cooperation under IS is sufficiently high, a given group of individuals prefer IS to FS. Let us define as "high contribution group" (respectively, "low contribution group") a group whose *initial* composition of individuals is such that $\hat{C} \geq \tilde{C}$ (respectively, $\hat{C} < \tilde{C}$).

[12] Note that, compared to the stage game in our experiment, we have slightly modified the timing of play to allow our one-period model to capture the relevant interactions occurring in the laboratory.

We describe a series of equilibria that arise when individuals can move between groups. Unlike the case in which individuals cannot move, it becomes important to distinguish between the different possible combinations of group "types" (either "high contribution" or "low contribution").

**Proposition 4** *Suppose individuals can move between groups. When both groups are "high contribution", under social preferences à la* (*Fehr and Schmidt* 1999), *the following equilibria exist*:

(a) *Both groups implement informal sanctions, no individual moves between groups, and all individuals contribute* $C_i^* \in [\tilde{C}, E]$, *for* $i = 1, ..., n^g$ *and* $g \in \{A, B\}$.
(b) *Groups implement different sanctioning institutions. Provided* (7) *holds after the moving stage, all individuals play under informal sanctions and contribute* $C^* \in [\tilde{C}, E]$.

**Proposition 5** *Suppose individuals can move between groups. Suppose also that groups have different types, that is, group A is* "*high contribution*", *whereas group B is* "*low contribution*". *Then, under social preferences à la* (*Fehr and Schmidt* 1999), *the following equilibria exist*:

(a) *The* "*high contribution*" *group A implements informal sanctions, whereas the* "*low contribution*" *group B implements formal sanctions. Individuals may or may not move from group B to group A—provided* (7) *holds in group A after moving has occurred—whereas no individual moves from A to B*.
(b) *The* "*high contribution*" *group A implements formal sanctions, whereas the* "*low contribution*" *group B implements informal sanctions. Either all individuals or some of them move from group B to group A*, *provided* (7) *holds in group B following the moving stage*.

*In all equilibria, individuals playing under formal sanctions* (*respectively, informal sanctions*) *contribute* $C^* = E \left( \text{respectively, } C^* \in [\tilde{C}, E] \right)$.

**Proposition 6** *Suppose individuals can move between groups. Suppose also that both groups are* "*low contribution*". *Then, under social preferences à la* (*Fehr and Schmidt* 1999), *the following equilibria exist*:

(a) *Both groups implement formal sanctions, either none or some individuals move between groups*, *and all individuals contribute* $C_i^* = E$, *for* $i = 1, ..., n^g$ *and* $g \in \{A, B\}$.
(b) *Groups implement different sanctioning institutions. Either all individuals or some of them move from the group implementing informal sanctions to the group*

*implementing formal sanctions, provided* (7) *holds in the former group following the moving stage*.

In all equilibria, individuals playing under formal sanctions (respectively, informal sanctions) contribute $C^* = E \left( \text{respectively, } C^* \in [\tilde{C}, E] \right)$.

Although the equilibria described in Propositions 4, 5 and 6 do not encompass all the possible equilibria arising in our dynamic game, they allow us to shed light on the interaction between individuals' ability to move between groups and institutional choices.[13] All in all, under selfish preferences, individuals' ability to move between groups does not fundamentally affect institutional choices or contributions (see Propositions 1 and 3). By contrast, under social preferences à la (Fehr and Schmidt 1999), the possibility for individuals to move between groups may affect their support for different institutions. In particular, whereas "high contribution groups" (respectively, "low contribution groups") always prefer IS (respectively, FS) in "closed" societies (see Proposition 2), Proposition 5 reveals that "high contribution groups" (respectively, "low contribution groups") may end up choosing FS (respectively, IS) when societies are "open". Intuitively, the expectation that individuals from a "low contribution group" might enter a "high contribution group" and destabilize its cooperative norms might lead the latter's *initial* members to implement more centralized forms of sanctioning institutions. Likewise, the expectation that some individuals might migrate from a "low contribution group" towards a group implementing FS (independently of whether it is "high contribution" or "low contribution"), might induce individuals in a "low contribution group" to implement IS in an attempt to improve cooperation under a different group size and composition.

In the empirical analyses reported in Sect. 3, we operationalize the notion of "high contribution group" (respectively "low contribution group") as a group in which the average contribution of its members is above (respectively, below) the median sample contribution—i.e., the median contribution of all the groups in the experiment—in the first five periods of each session, when institutions and group affiliations are fixed.[14]

Based on the intuitions from our theoretical framework, we state our main hypothesis:

**Hypothesis 1: voting**

(a) Subjects initially located in groups which exhibit above-median cooperation levels in the first five periods are more likely to vote in favor of informal sanctioning institutions when societies are "closed" than when they are "open".

[13] For instance, when groups *A* and *B* have different "types" (one is "high contribution" and the other is "low contribution"), it can be shown that there exist equilibria in which both groups implement the same type of sanctioning institutions. Although the study of these additional equilibria may be interesting in its own right, we believe that it would not significantly contribute to our understanding of the interaction between individuals' ability to move across "open" societies and support for sanctioning institutions.

[14] For robustness, we also assess the sensitivity of our results to alternative operationalizations of "high" and "low" contributions. See footnote 19 and the additional results reported in Appendix C.

(b) Subjects initially located in groups which exhibit below-median cooperation levels in the first five periods are more likely to vote in favor of informal sanctioning institutions when societies are "open" than when they are "closed".

Additionally, by comparing Propositions 1, 2 to Propositions 3, 5, 6, we have that formal sanctions perform equally well under any configuration of group members' preferences (i.e., selfish or social preferences) and independently of whether individuals can move between groups. As a consequence, because individuals always have the option of choosing formal sanctions, it must be the case that informal sanctions, whenever chosen, also perform equally well independently of whether individuals can move between groups.

Moreover, because individuals fully contribute to the public good under formal sanctions, the multiplicity of equilibria under informal sanctions implies that contributions should be somewhat lower under informal sanctions than under formal sanctions.

Finally, from Propositions 4, 5 and 6, when individuals are allowed to move between groups, there exist equilibria in which the size of the two groups differ. Intuitively, under informal sanctions, a change in the initial composition and size of the groups may be needed to achieve a sufficiently high level of contributions.

We can thus state the following additional hypotheses:

**Hypothesis 2: contributions**

(a) Under IS, contributions to the public good are similar across treatments. The same result holds under FS.
(b) Independently of whether subjects can move between groups, contributions to the public good are (weakly) higher under FS than under IS.

**Hypothesis 3: Moving** In "open" societies, subjects tend to locate unevenly between groups.

## 3 Results

We first compare subjects' voting behavior in our baseline and main treatments, and then examine the main determinants of subjects' contribution, punishment, and migration decisions.

### 3.1 Voting behavior

We start with an overview of group dynamics in voting behavior in the two treatments. Figure 1 plots the average percentage of people voting for IS over time in both MT and NMT.

The figure reveals that a larger fraction of the population chooses IS in all voting periods under NMT than under MT. As a result, the average percentage of subjects voting for IS across all periods and sessions is significantly larger in NMT (57.2%) than in MT (46.4%).[15]

To study whether voting depends on initial levels of cooperation in the IS regime, Fig. 2 divides groups depending on whether their average contribution to the public good in the first five periods of each session is above or below the median contribution of all groups in these initial periods. For both treatments, panel A plots the average share of subjects originally located in groups with initial contributions above the median who vote for IS. Panel B shows the average share of subjects originally located in groups with below-median initial contributions who vote for IS under the two treatments.

Figure 2, panel A reveals that participants whose group exhibited relatively high initial contribution levels (i.e., above the sample median) are more likely to vote for informal sanctioning institutions when the groups are closed than when they are open. The percentage of subjects who vote for IS in the last three voting stages is significantly larger in NMT (84%) than in MT (54%).[16,17] Panel B of the figure shows that, for participants whose group exhibited relatively low (below-median) initial contributions, the results reverse: subjects are less likely to vote for IS when the groups remain the same over the course of the experiment. Considering again only the last three voting periods, the corresponding proportions for NMT and MT are 32.7% and 46.0%, respectively.[18]

Result 1. Over the course of the experiment, a larger proportion of subjects who were originally located in a group with initial levels of cooperation above the median vote for informal sanctioning institutions (IS) in closed societies (NMT) than in open ones (MT). The opposite result holds for subjects who were originally located in a group with initial levels of cooperation below the median: in this case, a larger share of subjects vote

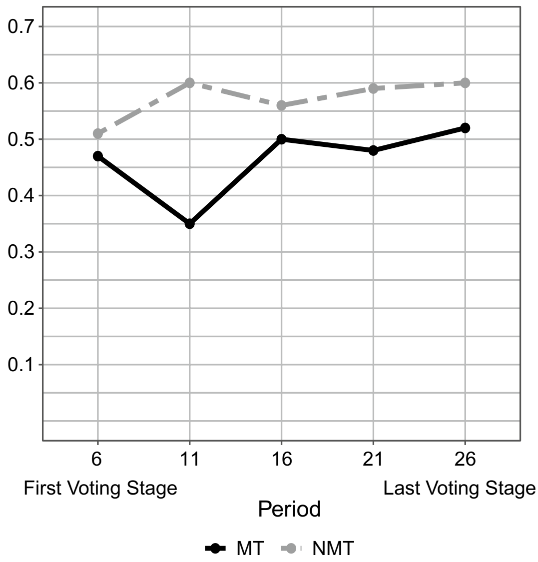
[15] $\chi^2(1) = 11.25$, $p = 0.00$, for a two-sided test of equality of proportions comparing the share of votes for informal sanctioning institutions—i.e., votes for IS as a percentage of votes cast in all voting periods—averaged across all NMT sessions against the corresponding average share across all MT sessions. A Mann–Whitney test taking each session as an independent observation—i.e., comparing the average proportion of votes for informal sanctions in each of the 10 NMT sessions against the average proportion of votes for IS in each of the 10 MT sessions—yields a similar conclusion: $z = 2.10$, $p = 0.03$, two-tailed.

[16] $\chi^2(1) = 30.17$, $p = 0.00$ for a two-tailed test for equality of proportions comparing the average share of votes for IS in the last 3 voting periods of all NMT sessions against the corresponding average share across all MT sessions. The same conclusion holds for a Mann-Whitney test taking each session as an independent observation: $z = 2.33$, $p = 0.02$, two-tailed.

[17] We choose the last three periods that include a voting stage to allow for some learning in the process. Results are the same if we take the last two periods that include a voting stage ($\chi^2(1) = 23.34$, $p = 0.00$, two-tailed equality of proportions test comparing the share of votes for IS across all NMT *versus* all MT sessions; Mann–Whitney test taking each session as an independent observation: $z = 2.28$, $p = 0.02$, two-tailed) or the last voting stage ($\chi^2(1) = 11.58$, $p = 0.00$; Mann–Whitney test: $z = 2.27$, $p = 0.02$).

[18] $\chi^2(1) = 5.04$, $p = 0.02$ for a two-sided equality of proportions test comparing the average share of votes for IS in the last 3 voting periods of all NMT sessions against the corresponding percentage averaged across all MT sessions. Mann-Whitney test taking each session as an independent observation: $z = 2.10$, $p = 0.03$, two-tailed.

**Fig. 1** Proportion of participants voting for IS over time, by treatment. Solid black (dashed gray) lines represent the proportion of subjects voting for IS under MT (NMT) in each of the periods in which a voting stage takes place, averaged across sessions



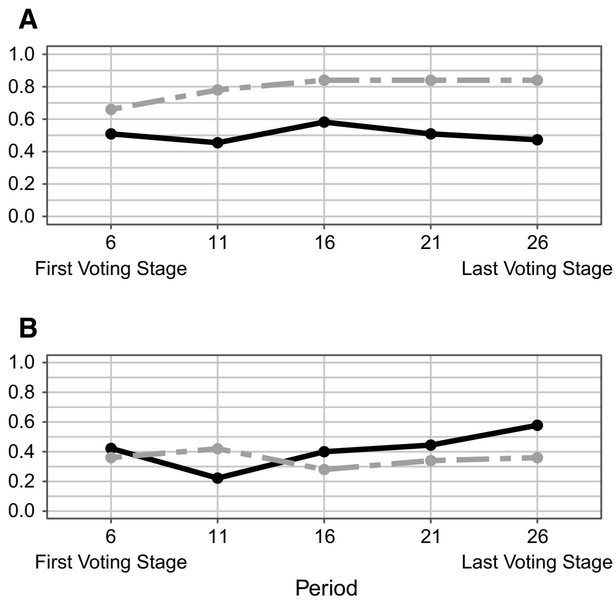




**Fig. 2** Proportion of participants voting for IS over time, by group's average initial contribution and treatment. For each period in which a voting stage takes place, panel **A**, **B** shows the proportion of subjects originally located in groups with above-median (below-median) initial contributions voting for IS under MT (solid black lines) and NMT (dashed gray lines), averaged across sessions

for informal sanctioning institutions (IS) in open societies (MT) than in closed ones (NMT).

Overall, *Result 1* is in line with our *Hypotheses 1.a* and *1.b*. Similar conclusions hold using alternative operationalizations of "high" and "low" initial group contributions, underscoring the robustness of this result.[19]

To better understand the mechanisms underlying the group-level patterns reported above, we next present individual-level analyses of voting decisions. The first column of Table 1 reports marginal effects (in percentage points) computed from a panel probit model fitted to data from the five periods that include a voting stage.[20] The dependent variable, $Vote_{i,t}$, equals 1 (0) if subject $i$ voted for (against) IS in period $t$. The explanatory variables include: $Moving_i$, a binary covariate that equals 1 if $i$ participated in MT, and 0 otherwise; $Contribution_{i,t-1}$-$Contribution_{g(-i),t-1}$, the difference between $i$'s contribution and the average contribution of the other members of her group in period $t-1$; $IS_{i,t-1}$, a dichotomous variable taking the value 1 if $i$ played under the informal sanctioning mechanism in the period preceding each voting stage, and 0 otherwise; and $Group\ Size_{g(i),t}$, the size of $i$'s group in period $t$. We also incorporate covariates capturing the behavior of the members of the two groups in the first five periods of the experiment: $High\ Initial\ Group\ Contribution_{g(i)}$, an indicator taking the value 1 when the average contribution in $i$'s group in those initial periods was greater than the median initial contribution of all the groups in the experiment, and 0 otherwise; the interaction between $High\ Initial\ Group\ Contribution_{g(i)}$ and $Moving_i$; $High\ Initial\ Group\ Payoff_{h \neq g}$, a dummy indicating whether the average payoff of the members of the other group—i.e., the group subject $i$ did not originally belong to—in the first 5 periods exceeded the median payoff of all the groups in those initial periods; and the interaction between $High\ Initial\ Group\ Payoff_{h \neq g}$ and $Moving_i$.[21]

In addition, the model includes subject-specific (correlated) random effects (Wooldridge 2005, 2010) to account for time-invariant individual heterogeneity, as well as period, group and session random intercepts controlling for temporal shocks affecting all subjects, for contemporaneous correlation between same-group

[19] We replicated the analysis using various absolute—rather than relative—criteria to operationalize "high" and "low" initial contributions. Specifically, we alternatively classify a group as having a "high" initial contribution if the average contribution of its members in the first five periods is: (i) more than half of the endowment (i.e., 25 tokens); (ii) more than 60% of the endowment (30 tokens); or (iii) more than 70% of the endowment (35 tokens). The findings summarized in *Result 1* are not sensitive to the particular operationalization used (see Fig. 6 and Table 5 in Appendix C).

[20] We report the "raw" parameter estimates in Table 6 of Appendix C. These must be transformed to obtain estimates for the marginal effects—i.e., the change in predicted probability associated with changes in the explanatory variables (Greene 2003, p. 667).

[21] As mentioned in Sect. 2, subjects receive information about the average contribution to the public good in their own group, but not in the other group. They are informed about the average payoffs in both groups, though. For completeness, we replicated the analyses replacing $High\ Initial\ Group\ Contribution_{g(i)}$ with $High\ Initial\ Group\ Payoff_{g(i)}$; the main findings remain unchanged (see Table 7 in the Appendix).

members, and for potential session effects, respectively (Poen 2009; Fréchette 2012).[22]

The estimates for *High Initial Group Contribution* $_{g(i)}$ imply that, in NMT, the tendency to vote for IS increases significantly when subjects experience relatively high initial group contributions. On average, the probability of voting for IS is about 37% points higher for individuals who belonged to groups that made contributions above the median at the beginning of the experiment than for subjects who were part of groups with below-median contributions in the first five periods. Note that informal sanctions are in place in both groups in these first five periods. Thus, this result suggests that participants are more willing to vote for IS when these institutions initially supported relatively high—i.e., above-median—contribution levels.

The negative and significant effect of the interaction between *High Initial Group Contribution*$_{g(i)}$ and $Moving_i$ shows that, in line with the results reported in Fig. 2 (see *Result 1*), participants originally located in groups with initial contributions above the median are more likely to vote for IS when they are in a society with exogenously fixed groups than when they are in societies with endogenous group formation. In fact, the positive relationship between *High Initial Group Contribution* $_{g(i)}$ and $Pr(Vote_{i,t} = 1)$ is exclusively driven by the voting behavior of subjects in NMT. In groups with initial contributions below the median, the mere possibility of moving between groups does not have a systematic influence on individuals' voting behavior. This is also observable from the statistically insignificant main effect of $Moving\ _i$ in the first row of Table 1.[23]

The probability of voting for informal sanctioning institutions also increases for subjects experiencing these institutions right before any given voting stage: the estimated marginal effect for $IS_{i,t-1}$ implies that subjects playing under informal institutions are almost 17% points more likely to vote for IS in the following period than those who experienced formal sanctions. The probability of voting for IS also decreases with the difference between the subject's contribution and the contribution of the other members of her group in the period immediately preceding a vote: each additional token a subject contributes above her group's average is associated with a 0.57% point decrease in $Pr(Vote_{i,t} = 1)$. In other words, people become more (less) willing to adopt formal (informal) sanctioning institutions when their contribution is high compared to the average contribution of their group.

The estimates in column (1) also indicate that the probability that a subject *i* votes for IS is not systematically influenced by the average payoff of the members of the

[22] As is well known, (bias-corrected) fixed effects estimators for panel probit models (e.g., Fernández-Val and Weidner 2016) cannot identify the coefficients of time-invariant covariates. For robustness, we also applied the estimator proposed by Kripfganz and Schwarz (2019) for fixed-effects linear panel models with time-invariant regressors. The results are aligned with those from the random effects probit models (see Table 8 in the Appendix).

[23] These results continue to hold when *High Initial Group Contribution*$_{g(i)}$ is defined as a dummy taking the value 1 when the average contribution in *i*'s group during the first 5 periods of the experiment is greater than a certain fraction—e.g., 50, 60 or 70%—of the endowment and 0 otherwise, instead of being measured with respect to the median initial contribution of all groups. See Table 9 in the Appendix.

**Table 1** Determinants of individual voting decisions

| | (1) | (2) | (3) |
|---|---|---|---|
| *Moving* $_i$ | 12.07 | 11.83 | 9.71 |
| | (9.67) | (8.48) | (7.68) |
| *High Initial Group Contribution* $_{g(i)}$ | 36.62*** | 29.45*** | 25.14*** |
| | (6.55) | (6.95) | (6.63) |
| *High Initial Group Contribution* $_{g(i)}$ × *Moving* $_i$ | – 31.24*** | – 26.13*** | – 21.64** |
| | (10.96) | (9.88) | (9.08) |
| *High Initial Group Payoff* $_{h \neq g}$ | 3.73 | 5.40 | 4.41 |
| | (6.61) | (5.57) | (5.20) |
| *High Initial Group Payoff* $_{h \neq g}$ × *Moving* $_i$ | – 10.03 | – 8.86 | – 7.22 |
| | (9.05) | (8.12) | (7.88) |
| $Contribution_{i,t-1} - Contribution_{g(-i),t-1}$ | – 0.57** | – 0.36 | – 0.18 |
| | (0.24) | (0.25) | (0.12) |
| $IS_{i,t-1}$ | 16.87*** | 9.77** | 11.60*** |
| | (4.19) | (4.23) | (4.39) |
| *Group Size* $_{g(i),t}$ | – 1.37 | – 0.85 | – 1.07 |
| | (0.97) | (0.88) | (0.87) |
| $Vote_{i,t-1}$ | | 28.39*** | 28.83*** |
| | | (4.80) | (4.74) |
| $Payoff_{i,t-1}$ | | | – 0.49** |
| | | | (0.19) |
| $Payoff_{i,t-1} \times IS_{i,t-1}$ | | | 2.39*** |
| | | | (0.48) |
| Observations | 1,000 | 800 | 800 |
| Log likelihood | – 549.47 | – 393.09 | – 378.89 |

The table reports the change in the probability of voting for informal institutions (in percentage points) associated with a change in the covariates; units of observation are individuals-per-period (with the sample restricted to the five periods that include a voting stage). All specifications include subject, period, group, and session random effects. Standard errors are presented in parentheses. Significance levels: *** at 1%, ** at 5%, * at 10%

other group in the first 5 periods under either treatment (MT or NMT).[24] Finally, the fact that the marginal effect of *Group Size* $_{g(i),t}$ on $Pr(Vote_{i,t} = 1)$ is statistically indistinguishable from zero is consistent with the results of Gürerk et al. (2006): an increase in the size of the group does not reduce the probability that participants support the implementation of informal sanctions.

To account for autocorrelation and for subjects' tendency to adjust their behavior over time (Smith 2013), column (2) in Table 1 adds the lag of the dependent

[24] We repeated the analysis replacing the *High Initial Group Payoff* $_{h \neq g}$ dummy with a continuous variable measuring the average payoff (in tokens) of the members of the other group in the first 5 periods of the experiment. The results remain unchanged, i.e. neither the average payoff of the members of group $h \neq g$ nor its interaction with $Moving_i$ has a statistically significant marginal effect on $Pr(Vote_{i,t} = 1)$.

variable to the regressors of the column (1) specification. The estimate for $Vote_{i,t-1}$ indicates that, holding everything else constant, the average propensity to support IS is more than 28% points higher for subjects who already voted in favor of that type of sanctions in the previous voting stage. This result suggests that subjects are consistent in their preferences over time. The estimates for most of the other explanatory variables are in line with those reported in column (1), the only exception being $Contribution_{i,t-1} - Contribution_{g(-i),t-1}$, which ceases to have a statistically significant effect on $Pr(Vote_{i,t} = 1)$ in this specification.

Column (3) adds $Payoff_{i,t-1}$, subject $i$'s payoff in the period previous to voting, along with its interaction with $IS_{i,t-1}$, to the set of explanatory variables of the column (2) specification. An increase of one token in the average subject's payoff in the period preceding a voting stage is associated with a 0.49 percentage point decrease in her willingness to vote for informal sanctions if such payoffs were generated under FS, and with a 2.39 percentage point increase if the earnings were generated under IS.

To further study whether members of groups with high initial levels of cooperation worry about inward migration of low contributors and therefore tend to vote for FS when societies are open, we now focus specifically on the first voting stage. Moreover, we consider only "heterogeneous sessions", i.e. sessions where the average initial contributions of one group were above the median sample value while the average initial contributions of the other group were below the median. There are 10 such sessions in our data (5 in MT and 5 in NMT), accounting for 50% of all the experimental sessions. We would expect the interaction between openness and initial average contributions to be particularly strong in these "heterogeneous sessions", since groups with above-median initial contributions in open societies face a high likelihood of inward migration by members of low-contribution groups. Additionally, initial contribution levels are expected to be most salient in the first period that includes a voting stage.

Figure 3 plots the percentage of subjects voting for IS in the first voting round of each treatment in the 10 "heterogeneous sessions", distinguishing, as in Fig. 2, between groups with initial contributions above and below the median. Since subjects are not allowed to move between groups before the first voting stage under either treatment, when migration is allowed (in MT), subjects' voting behavior is just affected by their expectation of out-members' future behavior (but not by the direct effect on $i$'s payoffs of out-members' past contributions *per se*).

The figure shows that the voting behavior in the two treatments is very similar for groups with initial contribution levels below the median: the percentage of subjects voting for IS in MT is 36.7%, compared to a 35% in NMT. The two proportions are statistically indistinguishable.[25]

[25] $\chi^2(1) = 0.01$, $p = 0.90$, for a two-sided equality of proportions test comparing the share of votes for IS averaged across all NMT sessions against the average share across all MT sessions. A Mann–Whitney test taking each "heterogeneous" session as an independent observation yields a similar conclusion: $z = 0.12$, $p = 1.00$, two-tailed.

Results however differ for groups with initial contributions above the median. In this case, 90% of subjects vote for IS when groups are fixed (NMT), whereas less than 47% vote for IS in MT—a statistically significant difference.[26] As expected, this difference is even larger than the corresponding effects reported in Fig. 2, panel A, for above-median contributors. These results support the view that subjects' expectations of other participants' behavior play a significant role in institutional choice.[27]

### 3.2 Contributions and punishment

Next, we analyze subjects' contribution to the public good and their punishment decisions in IS.[28] This analysis is key to better understand subjects' propensity to vote for different sanctioning institutions.

Figure 4 plots the evolution of the average individual contribution and the average punishment received by subjects when groups implement informal (panel A) and formal (panel B) sanctioning institutions under the two treatments.[29]

As seen in Fig. 4, panel A, when informal sanctioning institutions are in place, contributions are higher in NMT than in MT. In NMT, the average individual contribution across all sessions and periods is 44.69 tokens, whereas the corresponding value in MT is 37.46. This difference between treatments is statistically significant (Mann–Whitney test: $z = 3.89$, $p = 0.00$, two-tailed, taking each session as an independent observation).[30] This result seems to somehow contradict the results reported in Gürerk et al. (2014), where cooperation levels are found to be larger in open communities than in closed ones. Despite the apparent contradiction, there are substantial differences between our experimental design and that in Gürerk et al. (2014). Specifically, Gürerk et al. (2014) analyze a "voting with feet" system in which participants are not allowed to vote (cast ballots) on their preferred institutional setting. By contrast, our design allows participants to vote on the sanctioning institutions to be implemented in their group, regardless of whether groups are open or closed. Moreover, unlike Gürerk et al. (2014), our experiment allows for the implementation of a formal sanctioning scheme. All in all, the type of institutional self-selection we study seems to play an important role in explaining the different patterns of cooperation rates we observe in closed and open societies.

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[26] $\chi^2(1) = 9.11$, $p = 0.00$; Mann-Whitney: z=2.58, p=0.01.

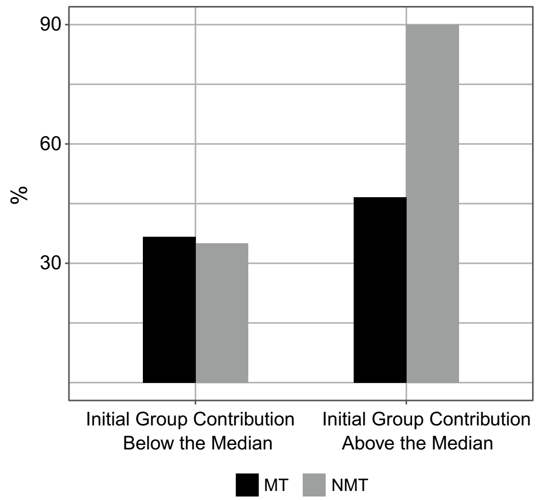
[27] Again, these results remain unchanged if the distinction between "high" and "low" group contributions is based on absolute measures. See Fig. 7 in Appendix C.

[28] For completeness, we also examine subjects' earnings, which are strongly related to their contributions and punishment decisions. These results are summarized in Appendix C (Fig. 8 and Table 10).

[29] Obviously, the fact that rule-sets are endogenous complicates the attribution of a causal relationship between the institutional setting and subjects' contribution and punishment decisions. We can nonetheless assess whether there is a systematic association between the type of institutions in place and the average contribution/punishment levels.

[30] This comparison considers only the average contributions from period 6 onward of each session. By contrast, average contributions are statistically indistinguishable between treatments in the first 5 periods of the experiment, when institutions and group affiliations are fixed (Mann–Whitney test: $z = 1.28$, $p = 0.20$, two-tailed, taking each session as an independent observation).

**Fig. 3** First vote on IS in "heterogeneous sessions". The figure plots the proportion of subjects voting for IS in groups with "high" (above-median) and "low" (below-median) average initial contributions in the first voting stage under both MT (black bars) and NMT (gray bars). The analysis is restricted to sessions in which the average initial contributions are above the median for one group and below the median for the other group




Under IS, we also observe a significant monotonic, positive trend in contributions in both MT and NMT (Mann–Kendall trend test: $z = 3.62, p = 0.00$ and $z = 3.46, p = 0.00$ for MT and NMT, respectively; both two-sided tests). Although the average contribution across sessions is also somewhat higher in NMT (45.12) than in MT (42.8) under FS, the difference between treatments is statistically indistinguishable from zero (Mann–Whitney test: $z = 0.65$, $p = 0.52$, two-tailed, taking each session as an independent observation). In this case, we observe that only contributions under NMT exhibit a significant monotonic trend (Mann–Kendall trend test: $z = 0.68, p = 0.50$, and $z = 3.76, p = 0.00$ for MT and NMT, respectively).

Altogether, Fig. 4 shows that the overall contribution levels are significantly larger under FS (43.60) than under IS (38.10) (Wilcoxon signed rank test: $z = 2.45$, $p = 0.01$, two-tailed).

Result 2. When informal sanctioning institutions are in place, contributions are higher in closed societies than in open ones. By contrast, when formal sanctioning institutions are adopted, societies are able to support high contributions to the public good in both treatments. Overall, contributions under formal sanctioning institutions are significantly larger than under informal ones.

Result 2 offers partial support for *Hypothesis 2.a* and full support for *Hypothesis 2.b*.[31]

[31] Arguably, the discrepancies between the results and *Hypothesis 2.a* could be explained by the fact that the repeated nature of the interactions in our experiment is not fully captured by the simplified one-period model on which our hypothesis is based.

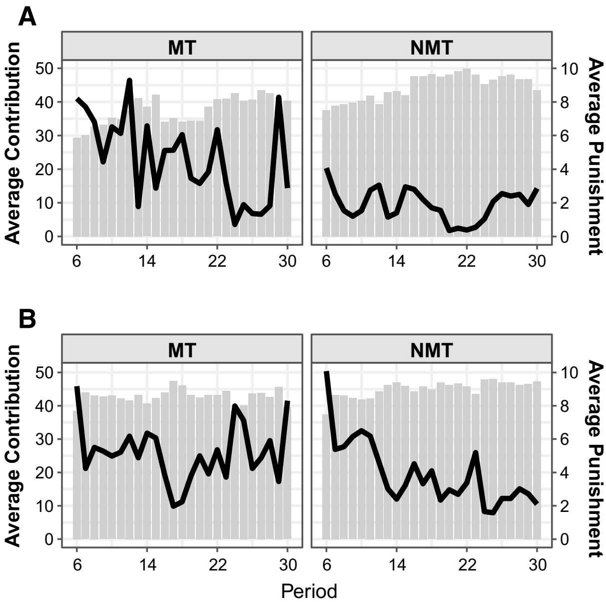



**Fig. 4** Average individual contribution and punishment received over time, by rule-set and treatment. **A**, **B** Plots the average contribution and punishment received by individuals located in groups implementing IS (FS) under the two treatments, in tokens. Bars represent the average contribution per period (measured on the left axis), while solid lines correspond to the average punishment received (measured on the right axis)

To study the main determinants of contributions, column (1) of Table 2 reports the marginal effects from a (doubly) censored regression model (Greene 2003) in which the dependent variable is $i$'s individual contribution in period $t$, $Contribution_{i,t}$.[32] The explanatory variables include some of the predictors used in the individual-level analyses of Table 1—$Moving_i$, $High\ Initial\ Group\ Contribution_{g(i)}$, $Contribution_{g(-i),t-1}$, $Group\ Size_{g(i),t}$, $IS_t$—along with the amount of punishment received by subject $i$ in the preceding period, a dummy indicating whether $i$ played under informal sanctions in period $t-1$, the interaction between $Punishment\ Received_{i,t-1}$ and $IS_{i,t-1}$, and the lag of the dependent variable (in order to control for persistence in individuals' contribution decisions). The model also incorporates subject, period, group and session random effects.[33]

Individual contributions to the public account increase significantly for subjects who originally belonged to a group with relatively "high"—i.e., above-median—initial contributions: each additional token contributed by $i$'s group in the first five

[32] "Raw" parameter estimates—representing the effect of the predictors on the uncensored latent variables—are reported in Appendix C (Table 11).

[33] The results are generally similar using the two-step method proposed by Honoré and Kesina (2017) to estimate fixed-effects censored regression models with time-invariant explanatory variables (Table 12 in Appendix C).

**Table 2** Determinants of individual contributions

| | (1) | (2) |
|---|---|---|
| $Moving_{\ i}$ | – 0.32 | – 0.76 |
| | (0.70) | (0.67) |
| $IS_{\ i,t}$ | – 3.41*** | − 3.33*** |
| | (0.45) | (0.45) |
| $Moving_i \times IS_{\ i,t}$ | – 1.27** | – 1.22** |
| | (0.51) | (0.50) |
| $High\ initial\ group\ contribution_{\ g(i)}$ | 1.29** | 1.01* |
| | (0.55) | (0.54) |
| $Contribution_{g(-i),t-1}$ | 0.21*** | 0.24*** |
| | (0.02) | (0.02) |
| $Group\ Size_{\ g(i),t}$ | 0.34*** | 0.16* |
| | (0.09) | (0.09) |
| $Contribution_{i,t-1}$ | 0.10*** | 0.11*** |
| | (0.02) | (0.02) |
| $Punishment\ received_{i,t-1}$ | – 0.03 | – 0.01 |
| | (0.03) | (0.02) |
| $IS_{\ i,t-1}$ | 1.81*** | 2.22*** |
| | (0.39) | (0.41) |
| $Punishment\ received_{i,t-1} \times IS_{\ i,t-1}$ | 0.05** | |
| | (0.02) | |
| $Punishment\ received_{i,t-1} \times \mathrm{IS}_{i,t-1} \times$ | | 0.12 |
| $\mathbb{1}\left(Contribution_{i,t-1} > Contribution_{g(-i),t-1}\right)$ | | (0.08) |
| $Punishment\ Received_{i,t-1} \times \mathrm{IS}_{i,t-1} \times$ | | 0.04** |
| $\mathbb{1}\left(Contribution_{i,t-1} < Contribution_{g(-i),t-1}\right)$ | | (0.02) |
| Observations | 4800 | 4800 |
| Log likelihood | – 11,558.33 | – 11,368.93 |

The table reports marginal effects for the covariates included in panel doubly-censored regression models for individual contributions. Units of observation are individuals-per-period. Both specifications include subject, period, group, and session random effects. Standard errors are presented in parentheses. Significance levels: *** at 1%, ** at 5%, * at 10%

periods of a session is associated with an increase of 1.29 tokens in the average subject's contribution at a later period $t$. All else equal, $i$'s contributions also increase with the contribution of the other members of her group in the previous period: the estimated marginal effect for $Contribution_{g(-i),t-1}$ implies that each additional token contributed by other subjects located in $i$'s group in the previous period leads to a subsequent increase in her own contribution of 0.21 tokens. These findings indicate that subjects' contribution behavior depends on their previous experience: individuals located—either at the beginning of the experiment or in a more recent period—in a group that cooperates more tend to contribute themselves more to the public good later on. The estimate for $Contribution_{i,t-1}$ reveals also a moderate but significant degree of persistence in individuals' propensity to cooperate: roughly 10% of subject $i$'s contribution in period $t$ is explained by her contribution in the previous period. Group size has a positive and significant effect on $Contribution_{i,t}$ as well.

On the contrary, $i$'s contribution in NMT declines by 3.41 tokens on average when her group is governed by IS, in line with *Result 2*. Furthermore, and also consistent with *Result 2*, the drop in contributions under informal sanctioning institutions is even more marked in open societies, as indicated by the negative and significant interaction between $Moving_i$ and $IS_{i,t}$.

Concerning the effect of punishment, the marginal effect of $Punishment\ Received_{i,t-1}$ on $i$'s contribution in period $t$ is statistically indistinguishable from zero in FS. By contrast, the estimate for the interaction between $Punishment\ Received_{i,t-1}$ and $\mathrm{IS}_{i,t-1}$ is positive and significant, indicating that punishment is on average more effective in raising contributions in IS than in FS,[34]

To account for the potential impact of "anti-social" punishment (i.e., punishment of high contributors) on contributions, column (2) re-estimates the relationship between $Punishment\ Received_{i,t-1}$ and $Contribution_{i,t}$ in IS, distinguishing between subjects who contributed more or less than their group mean.[35] The estimate for $Punishment\ Received_{i,t-1} \times \mathrm{IS}_{i,t-1} \times \mathbb{1}\left(Contribution_{i,t-1} < Contribution_{g(-i),t-1}\right)$ reveals a positive and significant relationship between the amount of informal punishment received in $t-1$ and subsequent contributions for individuals whose allocation to the group account is below the group average. By contrast, greater monetary sanctions do not have a systematic impact on $Contribution_{i,t}$ when $i$ contributed more than her fellow group members.

We end this section by investigating punishment behavior. We focus specifically on punishment under IS, since the relationship between contribution and punishment is straightforward in FS (see Sect. 2). Figure 4, panel A revealed that the average punishment received by subjects in groups implementing informal sanctioning institutions is lower in closed than in open societies—mainly because contributions are larger in NMT than in MT. The results presented in column (1) of Table 3, which reports marginal effects estimated from a random effects censored regression model examining the determinants of the amount of punishment received by subject $i$ in period $t$, are in line with this finding.[36] Holding individual contributions fixed, the punishment received by an average subject is systematically larger in MT than in NMT.

Column (2) expands the column (1) specification by including an interaction between $Moving_i$ and $i$'s contribution in $t$, along with the negative and positive deviation of $i$'s contribution from the average contribution of the other members of her group (measured in absolute values). Although the marginal effect of $Moving_i$ is no

[34] It is worth noting that, as shown in Fig. 4 average contributions are quite high in FS (and systematically higher than in IS). Hence, there is relatively "little room" for further increases. This may help explain the insignificant marginal effect of $Punishment\ Received_{i,t-1}$ on $Contribution_{i,t}$ in FS.

[35] Anti-social punishment may be present in IS (cf. e.g. Herrmann et al. 2008; Ertan et al. 2009), but is ruled out—by definition—in FS.

[36] "Raw" parameter estimates—representing the effect of the predictors on the uncensored latent variables—are reported in Appendix C (Table 13). The substantive results are similar using Honoré and Kesina (2017)'s fixed-effects estimator (Table 14).

longer significant, the estimate for $Contribution_{i,t} \times Moving_i$ indicates that, for any given contribution level, the amount of punishment received by the average subject is consistently higher in MT than in NMT. This provides some evidence that, in open societies, subjects may feel the need to be somewhat harsher in their punishment behavior, perhaps as a way to establish ground rules and discipline newcomers.

As for the impact of deviations from other group members' contributions, the positive and significant estimate for $max(Contribution_{g(-i),t}$—$Contribution_{i,t}$, 0) indicates that the punishment received by $i$ increases the further below the group average is her contribution. In contrast, the size of the positive deviation has no systematic impact on $i$' punishment, as the marginal effect of $max(Contribution_{i,t}$—$Contribution_{g(-i),t}$, 0) is statistically indistinguishable from zero. These findings underscore that there is more pro-social than anti-social punishment in IS, suggesting that informal punishment is mostly well-targeted at free riders.[37] Anti-social punishment is significantly more common in MT than in NMT: on average, there are 8.7 instances of anti-social punishment per session when moving is allowed, but only 2.7 under NMT (Mann-Whitney test: $z = 2.24$, $p = 0.02$, two-tailed, taking each session as an independent observation). This might be due to the changing composition of groups under MT, which could increase the occurrence of mis-coordination among group members and the likelihood that they resort to this type of punishment. The severity of anti-social punishment (i.e., the average number of tokens sent per anti-social punishment instance), however, is statistically indistinguishable across treatments.[38]

### 3.3 Migration behavior

We now analyze the main determinants of subjects' location choices in MT. This analysis sheds light on the endogenous dynamics of group formation. It also informs our results regarding the lack of significance of group size on subjects' voting behavior (Sect. 3.1). Groups size can vary more or less abruptly during the course of the experiment. Arguably, a "slow" rate of growth makes it easier for a host society implementing IS to discipline newcomers.[39] Hence, we expect that a "slow" increment in the number of subjects populating a given group will minimize any potentially negative impact of group size on the effectiveness of informal sanctioning institutions to induce high contributions. This, in fact, could explain why people do not vote for IS significantly less when the group size increases.

Figure 5, panel A plots the average share of subjects moving between groups in each period. A strictly positive fraction of subjects migrate in any given period and 20% on average switch groups each period over the course of the experiment, when moving is allowed. Panel B of the figure, in turn, distinguishes between moves

[37] This is also observed in Fig. 9 in Appendix C, which displays the punishment directed towards subjects in IS who contribute more than the group average, as a share of the total punishment in each period.

[38] The average number of tokens sent per instance of anti-social punishment is 9.07 in MT and 7 in NMT (Mann-Whitney test: $z = 0.34$, $p = 0.73$, two-tailed).

[39] Evidence pointing in this direction is provided by Weber (2006). The author finds that slow growth in group size has a positive effect on coordination.

**Table 3** Determinants of the amount of punishment received under IS

| | (1) | (2) |
|---|---|---|
| $Moving_i$ | 1.20** | 0.34 |
| | (0.50) | (0.51) |
| $Contribution_{i,t}$ | – 0.19*** | – 0.16*** |
| | (0.01) | (0.02) |
| $Contribution_{i,t} \times Moving_i$ | | 0.04** |
| | | (0.02) |
| $max(Contribution_{g(-i),t}$ - $Contribution_{i,t}, 0)$ | | 0.09*** |
| | | (0.02) |
| $max(Contribution_{i,t}$ - $Contribution_{g(-i),t}, 0)$ | | 0.01 |
| | | (0.01) |
| Observations | 2399 | 2399 |
| Log likelihood | – 4573.076 | – 4394.99 |

The table reports marginal effects for the covariates included in panel doubly-censored regression models examining the determinants of the amount of punishment received by subject *i* in period *t* under IS. Units of observation are individuals-per-period. Both specifications include subject, period, group, and session random effects. Standard errors are presented in parentheses. Significance levels: *** at 1%, ** at 5%, * at 10%

towards and moves away from IS when the two groups are implementing different rule-sets at the time those moves take place. Considering only periods/sessions in which one group implements IS while the other group has FS in place (42% of the observations), we find that 79% of the movements go from FS to IS, while only 21% of the group switches go in the opposite direction. These differences are statistically significant,[40,41]

Figure 5, panel C, which plots the average number of members in each group during the course of the experiment, indicates that subjects' migration patterns lead to groups diverging in size over time. This finding is in line with our *Hypothesis 3*. The average group size across all sessions and over the entire duration of the experiment is 7.03 for Group A and 2.97 for Group B, where, for ease of exposition, we label the group with the larger (smaller) number of members in each period of each session as Group A (Group B). In both cases, the average group size is significantly different

[40] $\chi^2(1)$=64.98, p=0.00, for a two-sided equality of proportions test comparing the number of moves from IS to FS to those in the other direction, as a proportion of the total number of group switches across all the periods and sessions in which the two groups implement different rule-sets. We also conducted a Wilcoxon signed rank test taking each of these (102) period-sessions as the unit of analysis; differences are again statistically significant: z = 8.63, p = 0.00.

[41] A possible explanation for this result is that IS is implemented because it functions relatively well. If this is the case, since IS is a relatively efficient institution with respect to FS, it tends to attract more subjects. Figure 10 in the Appendix complements this analysis by plotting the share of participants located in a group with IS in each period. This would be the consequence of the combination of voting and migration decisions.

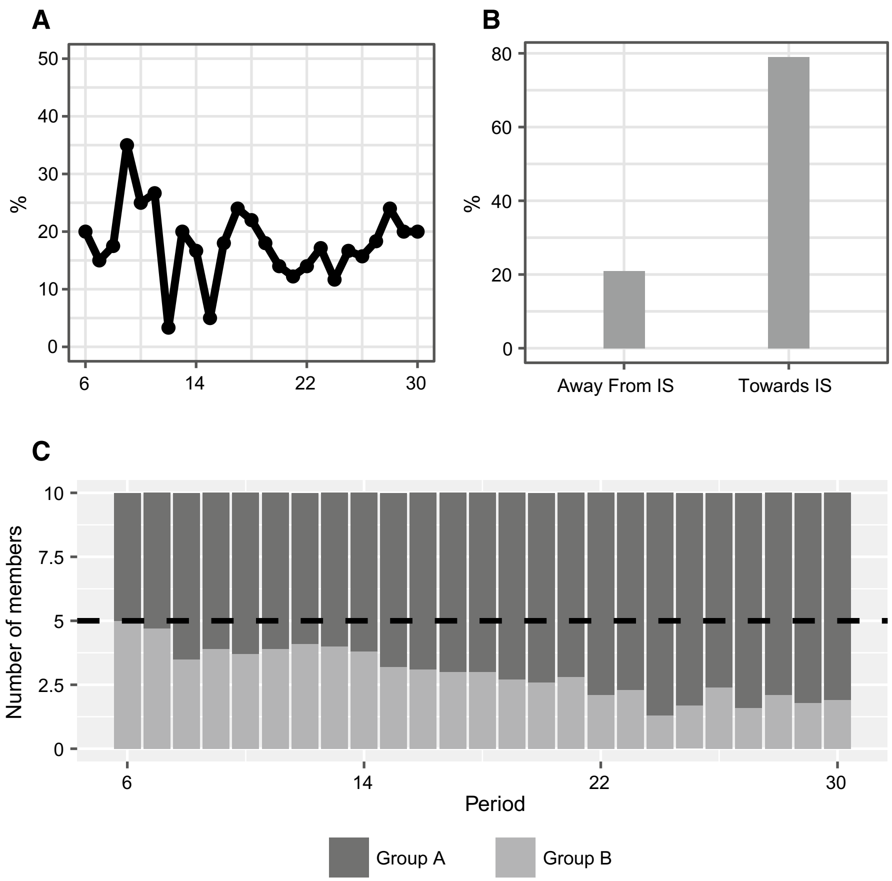



**Fig. 5** Migration patterns and group sizes over time. **A** Plots the average proportion of subjects moving between groups in each period (averaged across all sessions). **B** Plots the proportion of moves away from/towards groups that are implementing sanctions in the period in which the moves take place—out of the total number of movements between groups with different rule-sets in that period, across all sessions. Finally, **C** shows the number of members of Group A (the larger group, in darker bars) and Group B (the smaller group, in lighter bars) by period, averaged across sessions

from 5 (Mann-Whitney test: $z = 4.03$, $p = 0.00$, two-tailed, taking each session as an independent observation). The maximum group size in a given period (averaged across sessions) is 8.7. The mean per-period change in the number of members of a group—averaged across sessions—is 0.4. Hence, considering that moving is entirely costless, group size arguably grows (or shrinks) rather slowly in our experiment, which might help explain the insignificant marginal effect of $Group\ Size_{g(i),t}$ on $Pr(Vote_{i,t} = 1)$ reported in Table 1.

To analyze subjects' migration choices in greater detail, Table 4 reports marginal effects from a random effects panel probit model for migration decisions, in

**Table 4** Determinants of migration decisions

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| $IS_{\ i,t}$ | – 0.84 | – 0.84 | – 0.99 | – 0.98 |
| | (1.02) | (0.98) | (1.00) | (1.03) |
| $Contribution_{i,t} - Contribution_{g(-i),t}$ | 1.45** | 1.40** | 1.36** | 1.39** |
| | (0.70) | (0.67) | (0.66) | (0.69) |
| $Average\ Payoff_{h \neq g,t} - Average\ Payoff_{g,t}$ | 0.46** | 0.48** | 0.45** | 0.49** |
| | (0.19) | (0.19) | (0.19) | (0.20) |
| $Group\ Size_{\ g(i),t}$ | – 1.00** | – 1.00** | – 0.94** | – 0.96** |
| | (0.45) | (0.46) | (0.44) | (0.44) |
| $Vote\ Different\ from\ Group_{i,t-1}$ | 2.84* | 2.82* | 2.76* | 2.70* |
| | (1.46) | (1.49) | (1.46) | (1.49) |
| $Punishment\ Received_{i,t}$ | 0.21** | 0.20** | 0.19* | 0.19* |
| | (0.11) | (0.10) | (0.10) | (0.10) |
| $Punishment\ Received_{i,t} \times IS_{\ i,t}$ | – 0.01 | – 0.01 | – 0.01 | – 0.01 |
| | (0.08) | (0.08) | (0.08) | (0.08) |
| $Different\ Institutions_{\ t}$ | | – 0.26 | | – 0.33 |
| | | (0.91) | | (0.93) |
| $Migration_{\ i,t-1}$ | | | 1.58 | 1.52 |
| | | | (1.20) | (1.18) |
| Observations | 5000 | 5000 | 4800 | 4800 |
| Log likelihood | – 1000.38 | – 1000.32 | – 998.99 | – 998.92 |

The table reports the change in the probability of migrating (in percentage points) associated with a change in the covariates; units of observation are individuals-per-period. All the models include subject, period, group, and session random effects. Standard errors are presented in parentheses. Significance levels: *** at 1%, ** at 5%, * at 10%

percentage points.[42] The dependent variable is $Migration_{i,t}$, a dummy taking the value 1 if subject $i$ moves in period $t$, and 0 otherwise. The predictors in column (1) are essentially the same as those used in previous individual-level analyses, with the addition of: $Average\ Payoff_{h \neq g,t} - Average\ Payoff_{g,t}$, the difference between the average payoff of the members of the other group and the mean earnings of the subjects located in $i$'s group in period $t$; and $Vote\ Different\ from\ Group_{i,t-1}$, an indicator measuring whether $i$'s institutional choice in the most recent voting period differed from the decision of the majority of the members of her group.

Except for $IS_{i,t}$ and its interaction with $Punishment\ Received_{i,t}$, all the predictors included in the column (1) specification have a statistically significant effect on $Pr(Migration_{i,t} = 1)$. All else equal, each token increase in the difference between $i$'s contribution and the average contribution of the other members of her group is

[42] Parameter estimates are reported in Appendix C (Table 15). Since these specifications do not include time-invariant predictors, we also fitted two-way (subject and period) fixed-effects probit models using the method developed by Fernández-Val and Weidner (2016). The main results remain qualitatively similar (see Table 16).

associated with a 1.45 percentage point rise in the probability that she moves at the end of the period. Similarly, $Pr(Migration_{i,t} = 1)$ rises by 0.46 percentage points with each token increase in the difference between the average payoff of the members of the other group ($h$) and the average earnings of the members of $i$'s group ($g$). The probability that $i$ migrates also rises by 2.84 p. points when her group did not adopt $i$'s preferred rule-set in the most recent voting round.[43] Group size, by contrast, has a negative marginal effect on the probability that the average subject moves in a given period: each additional member of $i$'s group reduces $Pr(Migration_{i,t} = 1)$ by one percentage point. Finally, the higher the punishment received by $i$, the higher the probability she leaves the group, independently of whether the punishment originates from IS or FS.

Column (2) of Table 4 adds $Different\ Institutions_t$, an indicator for periods in which the two groups have different rule-sets in place, to the explanatory variables. Surprisingly, the probability of migration does not increase when the two groups implement different sanctioning institutions. Thus, it seems that subjects care more about institutions' effectiveness in generating high earnings than about the institutions *per se*. The estimated marginal effects for the other predictors remain similar to those presented in column (1).

Columns (3) and (4) of Table 4 present the results of two specifications that incorporate the lag of the dependent variable to the models of columns (1) and (2), respectively. All the substantive results discussed above remain unchanged. We also find no evidence of persistence in individuals' migration decisions: switching groups in the previous period does not have a significant effect on the propensity to migrate in the current period.

Result 3. On average, 20% of participants migrate every period, and group size varies by 0.4 members between any two consecutive periods. Subjects are not more likely to migrate when groups implement different sanctioning institutions, relative to periods in which the two groups play under the same rule-set. Yet, when sanctioning institutions do differ across groups, a vast majority of the "migrating subjects" moves towards groups that have informal sanctions in place.

## 4 Conclusions

The results of our experiment support the conclusion that the type of institutions emerging in a society depends on initial conditions as well as on whether the society is open or closed. When societies are sufficiently cooperative under an informal sanctioning mechanism, the presence of fixed groups reinforces the effectiveness of these institutions and, therefore, subjects' support for them. However, when moving in and out of groups is possible, it becomes harder to sustain cooperation. Importantly, when voting for their preferred institution, subjects consider both (*i*) current group members'

[43] For a similar result, see Cobo-Reyes et al. (2019).

behavior and (ii) potential newcomers' expected behavior. Overall, subjects' expectations play a key role in leading to an early adoption of centralized formal institutions. The implication is that allowing groups to form endogenously decreases subjects' willingness to adopt informal sanctioning institutions.

When a society is characterized by low initial cooperation levels under an informal sanctioning mechanism, the dynamics are essentially the opposite. In the presence of exogenously fixed groups, subjects can only rely on centralized formal institutions to improve cooperation. When groups form endogenously, however, self-selection (in the willingness to contribute and punish free-riders) can allow for an improvement in cooperation under informal sanctioning institutions, leading to a higher percentage of subjects voting for these institutions compared to the case of fixed groups.

Results in NMT confirm the findings in Markussen et al. (2014) that informal sanctions are surprisingly popular, even when competing against a formal sanctioning institution that is theoretically predicted to generate high levels of efficiency. Findings in the MT treatment, on the other hand, show that this popularity is fragile—it is significantly reduced when migration between groups is allowed. This rhymes well with results reported in Nicklisch et al. (2016) and Markussen et al. (2017), who show that the popularity of decentralized sanctions—relative to a centralized alternative—declines when the information provided to potential punishers about the contribution behavior of fellow group members is imperfect (i.e. cooperators are sometimes reported as free riders, and vice versa). The effect of opening group borders to migration is somewhat similar to providing imperfect information about other people's contributions. In both cases, it becomes more difficult to predict the future behavior of fellow group members, because information about their past behavior is imperfect. These studies all suggest that formal sanctioning institutions gain importance when networks of human interaction increase in complexity and fluidity, as would typically happen when a company expands, when a city grows, and when an economy develops an increasingly sophisticated division of labor. However, the present paper distinguishes itself from previous studies by pointing to the crucial impact of initial conditions. Groups with high initial levels of cooperation responded much more strongly to open borders than other groups.

The paper also contributes to the literature on the effects of endogenous choice of institutions. Whereas prior experimental work has compared exogenous and endogenous institutions in social dilemmas (e.g. Tyran and Feld 2006; Dal Bo et al. 2010), we compare an environment with one source of endogeneity (voting by ballot) to one with two sources of endogeneity (voting by ballot and voting with feet). We find lower levels of efficiency in the latter case than in the former, suggesting that the provision of multiple different avenues for people to affect their institutional environment may not necessarily lead to better outcomes.

# Appendix

## Appendix A: Experimental instructions

*Moving Treatment (MT)* You are now taking part in an economic experiment. Depending on your decisions and the decisions of other participants, you will be able to earn money. These instructions describe how you can earn money. Please read them carefully.

During the experiment you are not allowed to communicate with other participants. If you have a question, please raise your hand. One of us will come to answer your question. Sometimes you may have to wait a short while before the experiment continues. Please be patient. There will be a total of 30 periods in this experiment. We will explain carefully what you have to do.

*Allocations* In each period, you will be in a group of some size (you could also be by yourself). In each period, all members of your group, including you, will receive 50 points as "endowment".

You and the four others in your group simultaneously decide how to use the endowment. There are two possibilities:

1. Allocate points to a private account
2. Allocate points to a group account.

You will be asked to indicate the number of points you want to allocate to the group account. Only integers between 0 and 50 are allowed for this purpose. The remaining points will automatically be allocated to your private account.

Your payoff from the allocation decisions of yourself and the other members of your group will be as follows:

50 - (points you allocate to the group account) + $(1.6/n)\times$ (sum of points allocated by all members in the group to the group account)

where n is the number of members in your group. If you are the only member in a group, your payoff in that period will be 50 points.

EXAMPLE: Consider the case in which one participant belongs to a group in a particular period. Assume that the group has four members. For each point the participant allocates to her private account, she earns one point. For each point she allocates to the group account, she earns 0.4 points and each other group member also earns 0.4 points. If the total number of points in the group account is 100, the participant receives a payoff of $(100 \times 1.6)/4=40$ points from the group account (plus the number of points she has allocated to her\his private account).

**Groups:** In the first period you will be assigned to a group of five people (including yourself). You can be assigned to either GROUP A or GROUP B. Groups will remain fixed for the first 5 periods. Starting in period 6, you will decide at the end of each period whether you want to move to the other group. The other members of your group and the other group will also choose whether to move. We will now explain in more detail how this works.

**Payment rules:** there will be **two** different **Rule Sets**, which affect your earnings in different ways:

RULE SET 1 (peer to peer points reduction): In Rule Set 1, there are two stages in each period. In the first stage, you make your allocation decision as described above and learn the decisions of the other group members along with your earnings. In the second stage, you have an opportunity to reduce the earnings of others in your group at a cost to you. Here is how it works.

After the first stage of each period, you will be shown the amount allocated to the group account by each of the others in your group, in a random order, and in a box below that information you will be asked to enter a number of points (if any) that you wish to use to reduce the earnings of the individual who made that allocation decision (see below). Each point you allocate to reducing another's earnings reduces your own earnings by 1 point and reduces that individual's earnings by 3 points. Your own earnings can be reduced in the same way by the decisions of others in your group.

You are free to leave any or all of the others' earnings unchanged by entering 0 s in the relevant boxes. Your payoffs under Rule set 1 will be computed as follows:

50 - (points you allocate to group account) + $(1.6/n)\times$ (sum of points allocated by all in the group to the group account) − (points you spend to reduce others' earnings) − $3\times$ (sum of reduction points directed at you by others in your group)

EXAMPLE: Consider the case in which one participant belongs to a group of 5 people and RULE SET 1 (peer to peer points reduction) is in place. Suppose that you use 0 points to reduce the earnings of the first and second group members, you use 1 point to reduce the earnings of the third, and you use 2 points to reduce the earnings of the fourth. Suppose further that these individuals use 0, 1, 0 and 3 points, respectively, to reduce your earnings. Then, the third and fourth individuals' earnings for the period will be reduced by 3 and by 6 points, respectively, in addition to any reductions due to the decisions of others. Your own earnings for the period will be reduced by 3 points = the cost for you of imposing reductions on others, plus $(1\times 3)+(3\times 3)=12$ points = the reductions imposed on your earnings by others. At the end of the reduction stage, you will learn that your earnings were reduced by others by a total of 12 points, but you will not be told which individuals reduced your earnings or by how much any given individual reduced your earnings. Others will also not know who reduced their earnings.

The earnings reduction process is subject to two limits. First, you cannot assign more than 10 reduction points to any one individual in your group. Second, the total effective reduction of your earnings due to others' decisions in a given period cannot be greater than your total earnings from the allocation stage of that period. For example, if your earnings after the allocation stage are 26 points and if others use a total of 9 points to reduce your earnings, you will lose only 26 points, not $9\times 3=27$. However, the points that you spend to reduce the earnings of others are always costly to you, even if that brings your earnings for a period to less than zero. To continue with the example in which you earn 26 points before reductions and you lose 26 points due to the (27 points worth of) reductions others impose on you: if in the same period you have chosen to spend 3 points on reducing others' earnings, your total earnings for the period are - 3. Points lost in some periods are deducted from the £3 show-up fee that you will receive at the end of the experiment.

RULE SET 2 (automatic points reduction): In Rule Set 2, each individual pays a fixed fee of 5 points in each period. The fee is deducted from your earnings at the end of the period. In addition, each individual pays a fine equal to 80 percent of the amount of points she\he allocates to the private account in that period. Payoffs in each period are calculated as follows:

50 - (points you allocate to the group account) + $(1.6/n)\times$ (sum of points allocated by all members in the group to the group account) $-$ 0.8 $\times$ (points you allocate to the private account) $-$ 5

Note that if you are the only member in a group and Rule Set 2 applies, your payoff in that period will be 45 points (50 points minus the 5 points that you pay as fixed fee).

EXAMPLE: Consider the case in which one participant belongs to a group of 5 people and RULE SET 2 (automatic points reduction) is in place. If the participant contributes 20 points to the group account and keeps 30 points for the private account, given RULE SET 2, that participant would have to pay a fine of $0.8 \times 30 = 24$ points.

**Payment rules and groups in the first 5 periods:** If you are assigned to Group A, you will remain in Group A for the first 5 periods. If you are assigned to Group B you will stay in Group B for the first 5 periods. Rule set 1 (peer to peer points reductions) will apply in both groups for the first 5 periods.

**Information about earnings and rule sets:** At the end of each period, you will observe: (i) the average final earnings of your current group, (ii) the average final earnings of the other group, (iii) the rule set of your current group, and (iv) the rule set of the other group.

**Moving between groups:** Starting in period 6, at the end of each period you will decide whether you wish to leave the group you are currently in and move to the other group. The other members of your group, and the members of the other group, will make the same decision. You and all other participants are allowed to move back and forth between groups as many times as you wish. Hence, the size and member composition of the groups may change from period to period.

**Voting for rule sets:** Initial rules for Groups A and B will apply only for the first five periods. From period 6, every five periods you will have the chance to vote on whether you want Rule Set 1 or Rule Set 2 to be implemented in the group you currently belong to. Note that when you vote, you will vote to establish a rule in the group to which you belong in the voting period. Whichever rule set receives the highest number of votes will be in effect for the following five periods.

**Payment:** At the end of the experiment points will be converted into GBP at the rate of 15 points = 1GBP. You will be paid only for 3 periods, randomly chosen.

**SUMMARY**

1. There will be a total of 30 periods
2. You will begin by being in a group of five persons, including you.
3. In each period, you will allocate 50 points between a private and a group account.

4. Starting in period 6, you and everybody else can decide at the end of each period whether to move to the other group or not.
5. There are two Rule Sets. In Rule Set 1, each group member can reduce other group members' earnings after seeing the allocations of each individual to the group account. It costs 1 point to reduce the earnings of another group member by 3 points. In Rule Set 2, there is a fixed cost of 5 points in each period, deducted from your earnings at the end of the period. Each individual pays a fine equal to 80 percent of the amount of points he or she allocated to the private account. Every five periods you will vote on which of the two rule sets you prefer.
6. You will be paid only for 3 periods (randomly chosen).

**No Moving Treatment (NMT)** You are now taking part in an economic experiment. Depending on your decisions and the decisions of other participants, you will be able to earn money. These instructions describe how you can earn money. Please read them carefully.

During the experiment you are not allowed to communicate with other participants. If you have a question, please raise your hand. One of us will come to answer your question. Sometimes you may have to wait a short while before the experiment continues. Please be patient.

There will be a total of 30 periods in this experiment. We will explain carefully what you have to do.

**Allocations:** In each period, you will be in a group of 5 people. In each period, all members of your group, including you, will receive 50 points as "endowment".

You and the four others in your group simultaneously decide how to use the endowment. There are two possibilities:

1. Allocate points to a private account
2. Allocate points to a group account.

You will be asked to indicate the number of points you want to allocate to the group account. Only integers between 0 and 50 are allowed for this purpose. The remaining points will automatically be allocated to your private account.

Your payoff from the allocation decisions of yourself and the other members of your group will be as follows:

50 - (points you allocate to the group account) + $(1.6/5)\times$ (sum of points allocated by all members in the group to the group account)

EXAMPLE: Consider the case in which one participant belongs to a group in a particular period. Assume that the group has five members. For each point the participant allocates to her private account, she earns one point. For each point she allocates to the group account, she earns 0.32 points and each other group member also earns 0.32 points. If the total number of points in the group account is 100, the participant receives a payoff of $(100 \times 1.6)/5 = 32$ points from the group account (plus the number of points she has allocated to her\ his private account).

**Groups:** In the first period you will be assigned to a group of five people (including yourself). You can be assigned to either GROUP A or GROUP B. Your group will remain the same until the end of the experiment.

**Payment rules:** there will be **two** different **Rule Sets**, which affect your earnings in different ways:

RULE SET 1(peer to peer points reduction): In Rule Set 1, there are two stages in each period. In the first stage, you make your allocation decision as described above and learn the decisions of the other group members along with your earnings. In the second stage, you have an opportunity to reduce the earnings of others in your group at a cost to you. Here is how it works.

After the first stage of each period, you will be shown the amount allocated to the group account by each of the others in your group, in a random order, and in a box below that information you will be asked to enter a number of points (if any) that you wish to use to reduce the earnings of the individual who made that allocation decision (see below). Each point you allocate to reducing another's earnings reduces your own earnings by 1 point and reduces that individual's earnings by 2 points. Your own earnings can be reduced in the same way by the decisions of others in your group.

You are free to leave any or all of the others' earnings unchanged by entering 0 s in the relevant boxes. Your payoffs under Rule set 1 will be computed as follows:

50 - (points you allocate to group account) + $(1.6/5)\times$ (sum of points allocated by all in the group to the group account) – (points you spend to reduce others' earnings) – $2\times$ (sum of reduction points directed at you by others in your group)

EXAMPLE: Consider the case in which one participant belongs to a group of 5 people and RULE SET 1 (peer to peer points reduction) is in place. Suppose that you use 0 points to reduce the earnings of the first and second group members, you use 1 point to reduce the earnings of the third, and you use 2 points to reduce the earnings of the fourth. Suppose further that these individuals use 0, 1, 0 and 3 points, respectively, to reduce your earnings. Then, the third and fourth individuals' earnings for the period will be reduced by 2 and by 4 points, respectively, in addition to any reductions due to the decisions of others. Your own earnings for the period will be reduced by 3 points = the cost for you of imposing reductions on others, plus $(1\times2)+(3\times2)=8$ points = the reductions imposed on your earnings by others. At the end of the reduction stage, you will learn that your earnings were reduced by others by a total of 8 points, but you will not be told which individuals reduced your earnings or by how much any given individual reduced your earnings. Others will also not know who reduced their earnings.

The earnings reduction process is subject to two limits. First, you cannot assign more than 10 reduction points to any one individual in your group. Second, the total effective reduction of your earnings due to others' decisions in a given period cannot be greater than your total earnings from the allocation stage of that period. For example, if your earnings after the allocation stage are 26 points and if others use a total of 14 points to reduce your earnings, you will lose only 26 points, not $14\times2=28$. However, the points that you spend to reduce the earnings of others are always costly to you, even if that brings your earnings for a period to less than zero. To continue with the example in which you earn 26 points before reductions and you lose 26 points due to the (28 points worth of) reductions others impose on you: if in the same period you have chosen to spend 3 points on reducing others' earnings, your total earnings for the period are - 3. Points lost in some periods are deducted from the £3 show-up fee that you will receive at the end of the experiment.

RULE SET 2 (automatic points reduction): In Rule Set 2, each individual pays a fixed fee of 5 points in each period. The fee is deducted from your earnings at the end of the period. In addition, each individual pays a fine equal to 80 percent of the amount of points she he allocates to the private account in that period. Payoffs in each period are calculated as follows:

50 - (points you allocate to the group account) + $(1.6/5)\times$ (sum of points allocated by all members in the group to the group account) $-\ 0.8 \times$ (points you allocate to the private account) $-\ 5$

Note that if you are the only member in a group and Rule Set 2 applies, your payoff in that period will be 45 points (50 points minus the 5 points that you pay as fixed fee).

EXAMPLE: Consider the case in which one participant belongs to a group of 5 people and RULE SET 2 (automatic points reduction) is in place. If the participant contributes 20 points to the group account and keeps 30 points for the private account, given RULE SET 2, that participant would have to pay a fine of $0.8 \times 30 = 24$ points.

**Payment rules in the first 5 periods:** Rule set 1 (peer to peer points reductions) will apply in both groups for the first 5 periods.

**Information about earnings and rule sets:** At the end of each period, you will observe: (*i*) the average final earnings of your current group, (*ii*) the average final earnings of the other group, (*iii*) the rule set of your current group, and (*iv*) the rule set of the other group.

**Voting for rule sets:** Initial rules for Groups A and B will apply only for the first five periods. From period 6, every five periods you will have the chance to vote on whether you want Rule Set 1 or Rule Set 2 to be implemented in your group. Whichever rule set receives the highest number of votes will be in effect for the following five periods.

**Payment:** At the end of the experiment tokens will be converted into GBP at the rate of 15 points = 1GBP. You will be paid only for 3 periods randomly chosen.

**SUMMARY**

1. There will be a total of 30 periods
2. You will begin by being in a group of five persons, including you.
3. In each period, you will allocate 50 points between a private and a group account.
4. There are two Rule Sets. In Rule Set 1, each group member can reduce other group members' earnings after seeing the allocations of each individual to the group account. It costs 1 point to reduce the earnings of another group member by 2 points. In Rule Set 2, there is a fixed cost of 5 points in each period, deducted from your earnings at the end of the period. Each individual pays a fine equal to 80 percent of the amount of points he or she allocated to the private account. Every five periods you will vote on which of the two rule sets you prefer.
5. You will be paid only for 3 periods (randomly chosen).

## Appendix B: Proofs of Lemmas and Propositions

***Proof of Proposition 1*** Suppose first that IS are in place in group $g$, for $g \in \{A, B\}$. Then, for any given contribution profile $\{C_1, ..., C_{n^g}\}$ in group $g$, punishment does not occur at the punishment stage, that is, $R_{i,j} = 0$, for $i,j = 1, ..., n^g$, $i \neq j$ and $g \in \{A, B\}$. By backward induction, because $\frac{a}{n^g} < 1$ for all values $n^g > 1$, under IS we have $C_i = 0$, for $i = 1, ..., n^g$ and $g \in \{A, B\}$.

Suppose now that FS are in place in group $g$, for $g \in \{A, B\}$. Then, because sanctions are *deterrent*, all individuals contribute $C_i^* = E$ to the public good, for $i = 1, ..., n^g$ and $g \in \{A, B\}$. Finally, because $F < aE - E$, all individuals find it optimal to vote in favor of FS – i.e., $v_i^* = 0$, for $i = 1, ..., n^g$ and $g \in \{A, B\}$ – whenever they perceive the possibility of being pivotal. □

***Proof of Lemma 1*** The result directly follows from Proposition 5 in Fehr and Schmidt (1999). □

***Proof of Lemma 2*** The result directly follows from Proposition B1, point (*a*), as reported in the Appendix of Markussen et al. (2014). □

***Proof of Proposition 2*** Note that, from Lemmas 1 and 2, the equilibria of the two sub-games – that is, the sub-games starting after groups implement either IS or FS – involve homogeneous contributions and no inequality. From (6), it is easily seen that all individuals find it optimal to choose $v_i^* = 1$ (respectively, $v_i^* = 0$) if $\hat{C} \geq \tilde{C}$ (respectively, $\hat{C} < \tilde{C}$), whenever they perceive the possibility of being pivotal. □

***Proof of Proposition 3*** From the proof to Proposition 1, zero contributions to the public good occur under IS, whereas full contributions occur under FS. By backward induction, all individuals have an incentive to move to/remain in a group implementing FS and contribute $C_i^* = E$, for $i =, 1, ..., n$, provided at least one group implements these institutions. Finally, at the voting stage, multiple equilibria exist in which either (*i*) individuals in both groups vote in favor of FS or (*ii*) individuals in one group only vote in favor of FS (and all individuals move to/remain in this group). □

***Proof of Proposition 4*** Suppose both groups are "high contribution", that is, when no individuals move between groups, both groups support contributions $C_i \in \left[\tilde{C}, E\right]$, for $i = 1, ..., \tilde{n}^g$ and $g \in \{A, B\}$.

(a) Consider a putative equilibrium in which both groups implement IS, no individual moves between groups, and all individuals contribute $C_i^* = C^* \in \left[\tilde{C}, E\right]$, for $i = 1, ..., n^g, n^g = \tilde{n}^g$, and $g \in \{A.B\}$.

We solve the game by backward induction. At the *contribution stage*, because both groups are "high contribution", all individuals contribute $C_i^* = C^* \in \left[\tilde{C}, E\right]$, for $i = 1, ..., n^g, n^g = \tilde{n}^g$, and $g \in \{A.B\}$. At the *moving stage*, no individuals finds it profitable to move—i.e., $m_i^* = 0$, for $i = 1, ..., \tilde{n}^g$ and $g \in \{A.B\}$—provided she believes that the equilibrium level of contributions falls below $\tilde{C}$ were any individual to move between groups.[44] Finally, at the *voting stage*, given that group $B$ implements IS, group $A$'s *initial* members do not find it profitable to implement FS—i.e., $v_i^* = 1$, for $i = 1, ..., \tilde{n}^A$—as IS guarantee a level of contributions (weakly) higher than $\tilde{C}$. A similar reasoning holds for group $B$'s *initial* members. This proves point ($a$) in the Proposition.

(b) Consider a putative equilibrium in which group $A$ implements IS, whereas group $B$ implements FS. At equilibrium, group $A$'s *initial* members do not move—i.e., $m_i^* = 0$, for $i = 1, ..., \tilde{n}^A$, whereas group $B$'s *initial* members move to group $A$—i.e., $m_i^* = 1$, for $i = 1, ..., \tilde{n}^B$. All individuals contribute $C_i^* = C^* \in [\tilde{C}, E]$, for $i = 1, ..., n$.

We solve the game by backward induction. Suppose group $A$ implements IS, whereas group $B$ implements FS. Suppose also that all individuals are located in group $A$. At the *contribution stage*, provided (7) holds, contributions in group $A$ are such that $C_i^* = C^* \in [\tilde{C}, E]$, for $i = 1, ..., n$ (see Lemma 1). At the *moving stage*, from (4) and (6), no individual finds it profitable to unilaterally deviate by moving to/remain in group $B$ (and be the only individual located in group $B$). Finally, at the *voting stage*, given that group $B$ implements FS, group $A$'s *initial* members find it profitable to vote in favor of IS – i.e., $v_i^* = 1$, for $i = 1, ..., \tilde{n}^A$, as this leads to contributions (weakly) higher than $\tilde{C}$. Similarly, given that group $A$ implements IS, group $B$'s *initial* members find it profitable to vote in favor of FS – i.e., $v_i^* = 0$, for $i = 1, ..., \tilde{n}^B$, provided that they expect contributions to the public good to be no higher than $C^*$ under any possible individuals' equilibrium location choices of the sub-game in which both groups implement IS. This proves point ($b$) in the Proposition. □

***Proof of Proposition 5*** Suppose that group $A$ is "high contribution", that is, when no individuals move between groups, the *initial* members of group $A$ support contributions $C_i \in [\tilde{C}, E]$, for $i = 1, ..., n^A$. By contrast, suppose that group $B$ is "low contribution", that is, when no individuals move between groups, the *initial* members of group $B$ cannot support contributions $C_i \in [\tilde{C}, E]$, for $i = 1, ..., n^B$.

(a) In point (a), we first consider a putative equilibrium in which group $A$ implements IS and group $B$ implements FS, that is, $v_i^* = 1$, for $i = 1, ..., \tilde{n}^A$ (respectively, $v_i^* = 0$, for $i = 1, ..., \tilde{n}^B$). Also, in this equilibrium, individuals do not move between groups, i.e., $m_i^* = 0$ for $i = 1, ..., \tilde{n}^g$ and $g \in \{A, B\}$. Individuals located in group $A$ contribute $C_i^* \in [\tilde{C}, E]$, for $i = 1, ...n^A$, whereas individuals located in group $B$ contribute $C_i^* = E$, for $i = 1, ...n^B$.

44 This outcome may occur either because the conditions in Lemma 1—and in Proposition 5 in Fehr and Schmidt (1999)—are violated, or because individuals coordinate on an equilibrium involving relatively low contributions to the public good.

We solve the game by backward induction. Suppose group $A$ implements IS and group $B$ implements FS. When no individual moves between groups, we have $C_i^* \in [\tilde{C}, E]$, for $i = 1, ..., n^A$, and $C_i^* = E$, for $i = 1, ..., n^B$. Further, given individuals' behavior at the contribution stage, group $A$'s *initial* members do not find it profitable to move, as they expect $C_i \in [\tilde{C}, E]$, for $i = 1, ..., n^A$. Similarly, group $B$'s *initial* members do not find it profitable to move provided that, following their movement towards group $A$, they expect group $A$ to coordinate on $C_i < \tilde{C}$, for $i = 1, ..., n^A$. Given these out of equilibrium beliefs, individuals do not move from their *initial* groups – i.e., $m_i^* = 0$ for $i = 1, ..., \tilde{n}^g$ and $g \in \{A, B\}$. As a consequence, group $A$'s (respectively, group $B$'s) *initial* members find it profitable to vote in favor of IS (respectively, FS) – i.e., $v_i^* = 1$, for $i = 1, ..., \tilde{n}^A$ (respectively, $v_i^* = 0$, for $i = 1, ..., \tilde{n}^B$), whenever they perceive the possibility of being pivotal. This proves point ($a$) in the Proposition, for the case in which no individual moves between groups.

Consider now a putative equilibrium in which group $A$ implements IS and group $B$ implements FS, that is, $v_i^* = 1$, for $i = 1, ..., \tilde{n}^A$ (respectively, $v_i^* = 0$, for $i = 1, ..., \tilde{n}^B$). Also, in this equilibrium, one of group $B$'s *initial* members moves to group $A$, whereas all of group $A$'s *initial* members play $m_i^* = 0$, for $i = 1, ..., \tilde{n}^A$.[45] At equilibrium, individuals located in group $A$ contribute $C_i^* \in [\tilde{C}, E]$ (i.e., (7) holds), for $i = 1, ...n^A$, whereas individuals located in group $B$ contribute $C_i^* = E$, for $i = 1, ...n^B$.

We solve the game by backward induction. Suppose group $A$ implements IS and group $B$ implements FS. When only one individual moves from group $B$ to group $A$, provided (7) holds, we have $C_i^* \in [\tilde{C}, E]$, for $i = 1, ..., n^A$, and $C_i^* = E$, for $i = 1, ..., n^B$ (see Lemmas 1 and 2). At the *moving stage*, none of the group $A$'s *initial* members finds it profitable to move provided that she expects cooperation to break down in group $B$ following her movement. Similarly, only one of group $B$'s initial members finds it profitable to move to group $A$, as a movement by more than one individual is expected to generate a breakdown of cooperation (e.g., (7) does not hold for sufficiently high values of $n^A$). Finally, at the *voting stage*, given that group $B$ implements *FS*, group $A$'s *initial* members find it profitable to vote in favor of IS whenever they perceive a possibility of being pivotal. Similarly, given that group $A$ implements IS, group $B$'s *initial* members find it profitable to vote in favor of IS provided they expect not to be able to join a sufficiently cooperative group when both groups implement IS. For instance, this is the case when cooperation is expected to break down in both groups, when both groups implement IS. This proves point ($a$) in the Proposition, for the case in which some individuals move between groups.

(b) In point (b), we first consider a putative equilibrium in which group $A$ implements FS and group $B$ implements IS, that is, $v_i^* = 0$, for $i = 1, ..., \tilde{n}^A$ (respec-

[45] In this proof, we focus on the case in which only one individual moves from group $B$ to group $A$. It is straightforward to generalize the proof to the case in which more than one individual move from group $B$ to group $A$.

tively, $v_i^* = 1$, for $i = 1, ..., \tilde{n}^B$). Also, in this equilibrium, all individuals move to group $A$ – i.e., $m_i^* = 0$, for $i = 1, ..., \tilde{n}^A$, and $m_i^* = 1$, for $i = 1, ..., \tilde{n}^B$ – and contribute $C_i^* = E$, for $i = 1, ...n$.

We solve the game by backward induction. Suppose group $A$ implements FS and group $B$ implements IS. When all individuals are located in group $A$, from Lemma 2, we have $C_i^* = E$, for $i = 1, ..., n$. At the *moving stage*, from (4) and (6), no individual finds it profitable to unilaterally deviate by moving to/remaining in group $B$ (and be the only individual located in group $B$). Finally, at the *voting stage*, given that group $B$ implements IS, group $A$'s *initial* members find it profitable to vote in favor of FS, provided they expect coordination to break down when both groups implement IS. For instance, this occurs when any movement by group $A$'s *initial* members leads to no cooperation in both groups, and when a movement by one of group $B$'s initial members decreases cooperation in group $A$ in such a way that contributions in group $A$ fall below $\tilde{C}$, but group $A$ contributes more than what group $B$ would have contributed in the absence of movements between groups.[46] Similarly, given that group $A$ implements FS, group $B$'s initial members are indifferent between implementing IS and implementing FS – in which case all individuals contribute $C^* = E$ irrespective of their location. This proves point (*b*) in the Proposition, for the case in which all individuals move to the group implementing FS.

Consider now a putative equilibrium in which group $A$ implements FS and group $B$ implements IS, that is, $v_i^* = 0$, for $i = 1, ..., \tilde{n}^A$ (respectively, $v_i^* = 1$, for $i = 1, ..., \tilde{n}^B$). Also, in this equilibrium, group $A$'s *initial* members do not move – i.e., $m_i^* = 0$, for $i = 1, ..., \tilde{n}^A$, whereas one of group $B$'s initial members moves to group $A$. At equilibrium, individuals located in group $A$ (after the *moving stage* is over) contribute $C_i^* = E$, for $i = 1, ...n^A$. Individuals located in group $B$ (after the *moving stage* is over) contribute $C_i^* \in [\tilde{C}, E]$, for $i = 1, ...n^B$.

We solve the game by backward induction. Suppose group $A$ implements FS and group $B$ implements IS. From Lemma 2, equilibrium contributions in group $A$ are $C_i^* = E$, for $i = 1, ...n^A$. In group $B$, provided Lemma 1 holds, individuals coordinate on contributing $C_i^* \in [\tilde{C}, E]$, for $i = 1, ...n^B$.[47] At the *moving stage*, group $A$'s *initial* members do not move, provided out of equilibrium beliefs are such that any movement by group $A$'s *initial* members leads to a breakdown of cooperation in group $B$. Similarly, all but one of group $B$'s initial members have an incentive to remain in group $B$ given that $C_i^* \in [\tilde{C}, E]$ (respectively,

[46] It is easy to show that this behavior is an equilibrium of the sub-game in which both groups implement IS. To complete the description of the equilibrium, it is enough to add that, when both groups implement IS, a movement by more than one of group $B$'s *initial* members leads to a breakdown of cooperation in both groups.

[47] Recall that group $B$ is defined as "low contribution". We are therefore assuming that the decrease in the number of members leads to a new equilibrium in which cooperation is "high". Provided that the outgoing member is not a "conditionally cooperative enforcer", this statement is in line with Lemma 1 (see (7)).

$C_i^* \in [0, \tilde{C})\big)$ if only one member moves (respectively, if nobody moves). Finally, at the *voting stage*, given that group $B$ implements IS, group $A$'s *initial* members find it profitable to vote in favor of FS, provided they expect coordination to break down when both groups implement IS. For instance, this occurs when any movement by group $A$'s *initial* members leads to no cooperation in both groups, and when a movement by one of group $B$'s initial members decreases cooperation in group $A$ in such a way that contributions in group $A$ fall below $\tilde{C}$, but group $A$ contributes more than what group $B$ would have contributed absent movements between groups.[48] Similarly, given that group $A$ implements FS, group $B$'s initial members prefer to implement IS so that *initial* members who do not move enjoy contributions above $\tilde{C}$, whereas the *initial* member who moves to group $A$ enjoys $C^* = E$ under FS. All of group $B$'s *initial* members weakly prefer this outcome to one in which both groups implement FS and fully contribute to the public good. This proves point ($b$) in the Proposition, for the case in which some (but not all) individuals move to the group implementing FS. □

***Proof of Proposition 6*** Suppose both groups are "low contribution", that is, when no individuals move between groups, both groups support contributions $C_i \in [0, \tilde{C})$, for $i = 1, ..., n^g$ and $g \in \{A, B\}$.

(a) Consider a putative equilibrium in which both groups implement FS, no individual moves between groups, and all individuals contribute $C^* = E$.[49]

We solve the game by backward induction. At the *contribution stage*, from Lemma 2, all individuals contribute $C^* = E$ under FS. At the *moving stage*, since both groups implement FS, individuals are indifferent between moving or not. Finally, at the *voting stage*, provided that all individuals believe that the equilibrium level of contributions remains below $\tilde{C}$ under IS were any individual to move between groups, they all prefer to vote in favor of FS – i.e., $v_i^* = 0$, for $i = 1, ..., n$ – whenever they perceive the possibility of being pivotal. This proves point ($a$) in the Proposition.

(b) In point (b), we first consider a putative equilibrium in which group $A$ implements FS, whereas group $B$ implements IS. At equilibrium, all individuals are located in group $A$ and contribute $C^* = E$.

We solve the game by backward induction. At the *contribution stage*, from Lemma 2, all individuals contribute $C^* = E$. At the *moving stage*, from (4) and (6), no individual finds it profitable to unilaterally deviate by moving to/remain-

[48] It is easy to show that this behavior is an equilibrium of the sub-game in which both groups implement IS. To complete the description of the equilibrium behavior in this sub-game, it is enough to add that a movement by more than one of group $B$'s *initial* members leads to a breakdown of cooperation in both groups.

[49] As it will be clear from the proof, there exist equilibria in which both groups implement FS and individuals move between groups.

ing in group $B$ (and be the only individual located in group $B$). Finally, at the *voting stage*, given that group $B$ implements IS, group $A$'s *initial* members prefer to vote in favor of FS provided that (*i*) they expect sufficiently low levels of cooperation when both groups implement IS and (*ii*) they perceive the possibility of being pivotal. Similarly, given that group $A$ implements FS, group $B$'s *initial* members are indifferent between the two types of sanctioning institutions. This proves point (*b*) in the Proposition, for the case in which all individuals move to the group implementing FS.

Consider now a putative equilibrium in which group $A$ implements FS, whereas group $B$ implements IS. At equilibrium, group $A$'s *initial* members do not move – i.e., $m_i^* = 0$, for $i = 1, ..., \tilde{n}^A$, whereas only one of group $B$'s *initial* members moves to group $A$. Finally, $C_i^* = C^* = E$, for $i = 1, ..., n^A$, and $C_i^* = C^* \in [\tilde{C}, E]$, for $i = 1, ..., n^B$.

We proceed by backward induction. At the *contribution stage*, individuals located in group $A$ contribute $C^* = E$ (see Lemma 2), whereas individuals located in group $B$ coordinate on contributing $C^* \in [\tilde{C}, E]$ provided (7) holds (see Lemma 1). At the *moving stage*, none of group $A$'s *initial* members prefers to move to group $B$, provided they expect cooperation to break down (i.e., they expect $\hat{C} < \tilde{C}$) following such a movement. Similarly, only one of group $B$'s *initial* members prefers to move to group $A$, given that group $B$ is "low contribution", i.e., contributions in group $B$ are expected to be below $\tilde{C}$ in case of no movement. Finally, at the *voting stage*, given that group $B$ implements IS, group $A$'s initial members prefer to vote in favor of FS—i.e., $v_i^* = 0$, for $i = 1, ..., \tilde{n}^A$—provided they expect cooperation to break down when both groups implement IS. Likewise, given that group $A$ implements FS, group $B$'s *initial* members who plan to remain in (respectively, who plans to move from) group $B$ prefer(s) to vote in favor of IS (respectively, is indifferent between voting for FS and IS), as they expect contributions in group $B$ to be weakly higher than $\tilde{C}$ (respectively, as she expects to play under FS following the *moving stage*). This proves point (*b*) in the Proposition, for the case in which some (but not all) individuals move to the group implementing FS. □

## Appendix C: Additional results

See Figs. 6, 7, 8, 9, 10 and Tables 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15.

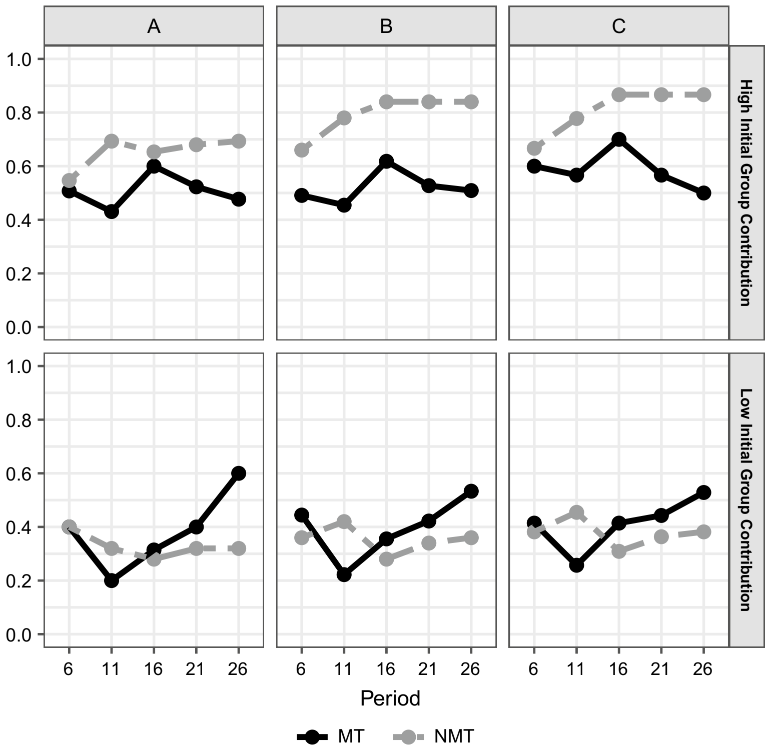



**Fig. 6** Impact of initial group contributions on voting patterns under alternative operationalizations of "high"/"low" contributions. The figure plots the percentage of subjects voting for IS over time under MT (solid black lines) and NMT (dashed gray lines), distinguishing groups by their average contribution in the first five periods of the experiment. The notion of "high contribution" group is operationalized using three distinct absolute measures. In panel A, a group is classified as having a "high" initial contribution when the average contribution of its members in the first five periods is more than half of the endowment (25 tokens). In panel B, a group is classified as having a "high" initial contribution when the average contribution of its members in the first five periods is more than 60% of the endowment (30 tokens). Finally, in panel C, a group is classified as having a "high" initial contribution when the average contribution of its members in the first five periods is more than 70% of the endowment (35 tokens). The patterns emerging from panels A–C are broadly similar to those displayed in Fig. 2. See also Table 5 below

**Table 5** Average proportion of subjects voting for IS under alternative definitions of "high contribution" group

| (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|
| Definition of "high" contribution | Group | Treatment | Proportion | Test for equality of proportions | Mann–Whitney test |
| More than half of the endowment | High contribution | MT | 53.33 | $\chi^2(1) = 8.29$ | $z = 1.78$ |
| | | NMT | 67.56 | $p = 0.00$ | $p = 0.08$ |
| | Low contribution | MT | 43.81 | $\chi^2(1) = 4.20$ | $z = 1.95$ |
| | | NMT | 30.67 | $p = 0.04$ | $p = 0.06$ |
| More than 60% of the endowment | High contribution | MT | 55.15 | $\chi^2(1) = 29.18$ | $z = 2.32$ |
| | | NMT | 84.00 | $p = 0.00$ | $p = 0.02$ |
| | Low contribution | MT | 43.70 | $\chi^2(1) = 3.22$ | $z = 1.95$ |
| | | NMT | 32.67 | $p = 0.07$ | $p = 0.06$ |
| More than 70% of the endowment | High contribution | MT | 58.89 | $\chi^2(1) = 21.08$ | $z = 2.29$ |
| | | NMT | 86.67 | $p = 0.00$ | $p = 0.02$ |
| | Low contribution | MT | 46.19 | $\chi^2(1) = 3.15$ | $z = 1.81$ |
| | | NMT | 35.15 | $p = 0.07$ | $p = 0.08$ |

This table replicates the analysis presented in Sect. 3.1 (Fig. 2), using alternative absolute - rather than relative - criteria to distinguish between "high contribution" and "low contribution" groups. Specifically, as seen in column (1), a group is alternatively classified as having a "high" initial contribution if the average contribution of its members in the first five periods is: (i) more than half of the endowment (25 tokens); (ii) more than 60% of the endowment (30 tokens); or (iii) more than 70% of the endowment (35 tokens). For each of these operationalizations, column (4) reports the average share of subjects originally located in "high contribution" and "low contribution" groups who vote for IS in the last three voting stages under MT and NMT. For every group "type" (i.e., "high contribution" or "low contribution"), column (5) reports test statistics and p-values from two-tailed equality of proportions tests comparing the average shares across all the sessions under MT and all the sessions under NMT. Column (6), in turn, reports test statistics and p-values from Mann–Whitney tests taking each session as an independent observation. As shown in columns (4)–(6), the substantive findings summarized in *Result 1* continue to hold when these alternative measures are used. In all cases, we observe that a larger proportion of subjects who were originally located in a group with high initial levels of cooperation vote for informal sanctioning institutions in NMT than in MT. The opposite holds for subjects who were originally located in a group with low initial levels of cooperation

**Table 6** Parameter estimates Panel probit models for voting

| | (1) | (2) | (3) |
|---|---|---|---|
| *Intercept* | − 0.54 | − 0.99*** | − 0.89*** |
| | (0.35) | (0.32) | (0.30) |
| $Moving_{\ i}$ | 0.49 | 0.49 | 0.41 |
| | (0.39) | (0.35) | (0.32) |
| $High\ Initial\ Group\ Contribution_{\ g(i)}$ | 1.39*** | 1.12*** | 1.00*** |
| | (0.30) | (0.30) | (0.28) |
| $High\ Initial\ Group\ Contribution_{\ g(i)} \times Moving_{\ i}$ | − 1.10*** | − 0.94** | − 0.82** |
| | (0.38) | (0.35) | (0.34) |
| $High\ Initial\ Group\ Payoff_{\ h \neq g}$ | 0.16 | 0.23 | 0.19 |
| | (0.27) | (0.24) | (0.23) |
| $High\ Initial\ Group\ Payoff_{\ h \neq g} \times Moving_{\ i}$ | − 0.41 | − 0.36 | − 0.29 |
| | (0.33) | (0.33) | (0.32) |
| $Contribution_{i,t-1} - Contribution_{\ g(-i),t-1}$ | − 0.25** | − 0.17 | − 0.18 |
| | (0.10) | (0.12) | (0.12) |
| $IS_{\ i,t-1}$ | 0.57*** | 0.35** | 0.43*** |
| | (0.14) | (0.15) | (0.15) |
| $Group\ Size_{\ g(i),t}$ | − 0.05 | − 0.03 | − 0.04 |
| | (0.03) | (0.03) | (0.03) |
| $Vote_{i,t-1}$ | | 0.91*** | 0.99*** |
| | | (0.14) | (0.14) |
| $Payoff_{i,t-1}$ | | | − 0.66** |
| | | | (0.26) |
| $Payoff_{i,t-1} \times IS_{\ i,t-1}$ | | | 1.81*** |
| | | | (0.37) |
| Observations | 1000 | 800 | 800 |
| Log likelihood | − 549.47 | − 393.09 | − 378.89 |

The table reports "raw" parameter estimates for the panel probit models used to compute the marginal effects presented in Table 1. Units of observation are individuals-per-period, with the sample restricted to the five periods that include a voting stage. All specifications include subject, period, group, and session random effects. Standard errors are presented in parentheses. Significance levels: *** at 1%, ** at 5%, * at 10%

**Table 7** Models for voting replacing *High Initial Group Contribution*$_{g(i)}$ with *High Initial Group Payoff*$_{g(i)}$

| | (1) | (2) | (3) |
|---|---|---|---|
| *Moving* $_i$ | 11.59 | 13.17 | 10.90 |
| | (10.12) | (9.43) | (8.09) |
| *High Initial Group Payoff* $_{g(i)}$ | 33.56*** | 25.79*** | 23.70*** |
| | (6.66) | (6.85) | (6.28) |
| *High Initial Group Payoff* $_{g(i)} \times$ *Moving* $_i$ | − 19.36* | − 19.76* | − 16.40* |
| | (11.33) | (10.28) | (9.03) |
| *High Initial Group Payoff* $_{h \neq g}$ | 8.77 | 10.49 | 9.17 |
| | (6.83) | (6.50) | (5.78) |
| *High Initial Group Payoff* $_{h \neq g} \times$ *Moving* $_i$ | − 13.39 | − 11.15 | − 9.18 |
| | (10.13) | (9.74) | (8.80) |
| $Contribution_{i,t-1} - Contribution_{g(-i),t-1}$ | − 0.56** | − 0.36 | − 0.18 |
| | (0.24) | (0.24) | (0.12) |
| $IS_{i,t-1}$ | 18.19*** | 10.91** | 12.20*** |
| | (4.17) | (4.26) | (4.39) |
| *Group Size* $_{g(i),t}$ | − 1.47 | − 0.93 | − 1.20 |
| | (0.97) | (0.89) | (0.88) |
| $Vote_{i,t-1}$ | | 28.42*** | 28.70*** |
| | | (4.85) | (4.74) |
| $Payoff_{i,t-1}$ | | | − 0.52*** |
| | | | (0.19) |
| $Payoff_{i,t-1} \times IS_{i,t-1}$ | | | 2.54*** |
| | | | (0.47) |
| Observations | 1,000 | 800 | 800 |
| Log likelihood | − 550.96 | − 395.27 | − 379.02 |

The table replicates the analyses in Table 1 of the paper, replacing *High Initial Group Contribution*$_{g(i)}$ with *High Initial Group Payoff*$_{g(i)}$ as a predictor of $Pr(Vote_{i,t} = 1)$. Units of observation are individuals-per-period, with the sample restricted to the five periods that include a voting stage. All specifications include subject, period, group, and session random effects. Standard errors are presented in parentheses. Significance levels: *** at 1%, ** at 5%, * at 10%

**Table 8** Parameter estimates fixed effects models for voting

| | (1) | (2) | (3) |
|---|---|---|---|
| *Intercept* | 0.26*** | 0.18*** | 0.21*** |
| | (0.06) | (0.06) | (0.06) |
| $Moving_{\ i}$ | 0.04 | 0.03 | 0.03 |
| | (0.04) | (0.04) | (0.04) |
| $High\ Initial\ Group\ Payoff_{\ g(i)}$ | 0.28*** | 0.14*** | 0.11** |
| | (0.04) | (0.04) | (0.04) |
| $High\ Initial\ Group\ Payoff_{\ g(i)} \times Moving_{\ i}$ | − 0.20*** | − 0.11* | − 0.10* |
| | (0.06) | (0.06) | (0.06) |
| $High\ Initial\ Group\ Payoff_{\ h \neq g}$ | 0.04 | 0.07 | 0.06 |
| | (0.05) | (0.05) | (0.05) |
| $High\ Initial\ Group\ Payoff_{\ h \neq g} \times Moving_{\ i}$ | − 0.03 | − 0.04 | − 0.03 |
| | (0.05) | (0.05) | (0.05) |
| $Contribution_{\ i,t-1} - \text{Contribution}_{g(-i),t-1}$ | − 0.37*** | − 0.37*** | − 0.35*** |
| | (0.12) | (0.12) | (0.12) |
| $IS_{\ i,t-1}$ | 0.33*** | 0.33*** | 0.32*** |
| | (0.03) | (0.03) | (0.03) |
| $Group\ Size_{\ g(i),t}$ | − 0.01 | − 0.01 | − 0.01 |
| | (0.01) | (0.01) | (0.01) |
| $Vote_{i,t-1}$ | | 0.31*** | 0.31*** |
| | | (0.03) | (0.03) |
| $Payoff_{i,t-1}$ | | | − 0.28*** |
| | | | (0.07) |
| $Payoff_{\ i,t-1} \times IS_{\ i,t-1}$ | | | 0.60*** |
| | | | (0.09) |
| Observations | 1000 | 800 | 800 |

The table reports parameter estimates for (linear) fixed-effects panel models for voting obtained using the two-step approach proposed by Kripfganz and Schwarz (2019) to identify the coefficients of time-invariant regressors. Units of observation are individuals-per-period, with the sample restricted to the five periods that include a voting stage. Standard errors are presented in parentheses. Significance levels: *** at 1%, ** at 5%, * at 10%

**Table 9** Panel probit models for voting using alternative definitions of "high contribution" group

| | (1) | (2) | (3) |
|---|---|---|---|
| *Intercept* | − 0.43 | − 0.54 | − 0.54 |
| | (0.41) | (0.35) | (0.35) |
| $Moving_{\ i}$ | 0.28 | 0.37 | 0.38 |
| | (0.47) | (0.39) | (0.38) |
| $High\ Initial\ Group\ Contribution_{\ g(i)}$ | 1.10*** | 1.39*** | 1.39*** |
| | (0.33) | (0.30) | (0.30) |
| $High\ Initial\ Group\ Contribution_{\ g(i)} \times Moving_{\ i}$ | − 0.73* | − 0.94** | − 0.89** |
| | (0.42) | (0.38) | (0.38) |
| $High\ Initial\ Group\ Payoff_{\ h \neq g}$ | 0.14 | 0.16 | 0.15 |
| | (0.27) | (0.26) | (0.26) |
| $High\ Initial\ Group\ Payoff_{\ h \neq g} \times Moving_{\ i}$ | − 0.26 | − 0.56 | − 0.47 |
| | (0.38) | (0.36) | (0.37) |
| $Contribution_{i,t-1} - Contribution_{\ g(-i),t-1}$ | − 0.26** | − 0.26** | − 0.26** |
| | (0.10) | (0.10) | (0.10) |
| $IS_{\ i,t-1}$ | 0.61*** | 0.57*** | 0.57*** |
| | (0.14) | (0.14) | (0.14) |
| $Group\ Size_{\ g(i),t}$ | − 0.05 | − 0.05 | − 0.05 |
| | (0.03) | (0.03) | (0.03) |
| Observations | 1,000 | 1,000 | 1,000 |
| Log likelihood | − 553.86 | − 548.58 | − 548.16 |

The table reports parameter estimates for the baseline panel probit model of voting (Tables 1 and 6, column 1), but using absolute—rather than relative—measures to define the $High\ Initial\ Group\ Contribution_{g(i)}$ indicator. Specifically, following the sames criteria as in Fig. 6 and Table 5, $High\ Initial\ Group\ Contribution_{\ g(i)}$ takes the value 1 when the average contribution in *i*'s group in the first five periods of the experiment is: more than half of the endowment (column 1); more than 60% of the endowment (column 2); or more than 70% of the endowment (column 3). Units of observation are individuals-per-period, with the sample restricted to the five periods that include a voting stage. All specifications include subject, period, group, and session random effects. Standard errors are presented in parentheses. Significance levels: *** at 1%, ** at 5%, * at 10%

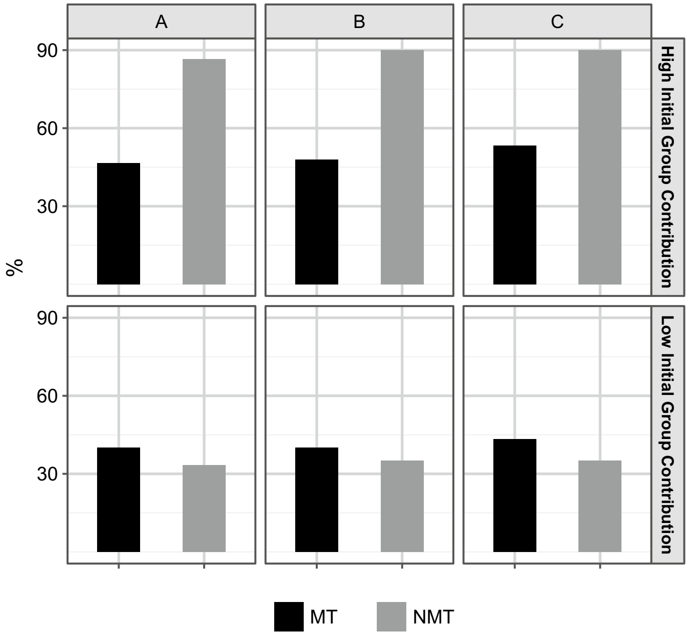



**Fig. 7** First vote on IS in "heterogeneous sessions", using alternative definitions of "high contribution" group. The figure replicates the analysis summarized in Fig. 3, but using absolute measures to distinguish between "high contribution" and "low contribution" groups. In panel A, a group is classified as having a "high" initial contribution when the average contribution of its members in the first five periods is more than half of the endowment (25 tokens). In panel B, a group is classified as having a "high" initial contribution when the average contribution of its members in the first five periods is more than 60% of the endowment (30 tokens). Finally, in panel C, a group is classified as having a "high" initial contribution when the average contribution of its members in the first five periods is more than 70% of the endowment (35 tokens). Regardless of the particular operationalization of "high"/"low" contributions, the main patterns emerging from the figure are similar to those displayed in Fig. 3. The proportion of subjects originally located in "low contribution" groups who vote for IS in the first voting round of the heterogeneous sessions is somewhat higher under MT than under NMT. Among subjects originally located in "high contribution" groups, however, the share of votes for informal sanctions is much higher under NMT than under MT

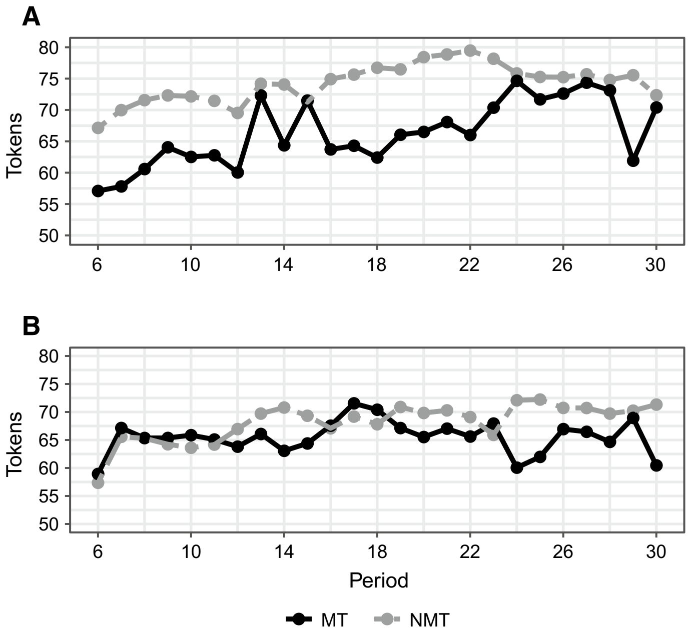



**Fig. 8 Average individual payoffs per-period over time, by rule-set and treatment.** The figure complements the information displayed in Fig. 4, plotting the average individual payoff per-period in groups implementing informal (panel A) and formal (panel B) sanctioning institutions under both treatments. Solid black (dashed gray) lines represent the average individual payoff per-period under MT (NMT), averaged across sessions. As in the case of contributions, the average payoff under IS is significantly higher in NMT (74.41 tokens) than in MT (66.86 tokens) (Mann-Whitney test: $z = 2.04$, $p = 0.04$, two-tailed, taking each session as an independent observation). By contrast, differences between treatments are not statistically significant under FS (Mann-Whitney test: $z = 0.98$, $p = 0.33$, two-tailed, taking each session as an independent observation)

**Table 10** Difference in average individual payoffs between rule-sets, by treatment

| (1) Treatment | (2) Rule-set | (3) Average individual payoff (per-period, in tokens) | (4) Wilcoxon signed rank test (two-sided) |
|---|---|---|---|
| MT | IS | 60.63 | |
| | FS | 65.59 | $z = -2.29$, $p = 0.02$ |
| NMT | IS | 67.59 | |
| | FS | 68.29 | $z = -0.41$, $p = 0.74$ |

For every combination of treatment (column 1) and rule-set (column 2), column (3) gives the average individual payoff per-period across all the sessions in which that combination is in place. As in the case of contributions, we observe that differences in payoffs are driven by the behavior of subjects in open societies: while in MT the average individual payoff per-period is significantly higher under FS than under IS, the difference between the two institutional settings is statistically indistinguishable from zero in NMT (column 4)

**Table 11** Parameter estimates Censored regression models for contributions

| | (1) | (2) |
|---|---|---|
| *Intercept* | 7.77*** | 3.00 |
| | (2.50) | (2.56) |
| $Moving_i$ | − 1.28 | − 3.03 |
| | (2.82) | (2.67) |
| $IS_{i,t}$ | − 13.66*** | − 13.32*** |
| | (1.79) | (1.78) |
| $Moving_i \times IS_{i,t}$ | − 5.08** | − 4.90** |
| | (2.04) | (2.00) |
| $High\ Initial\ Group\ Contribution_{g(i)}$ | 5.18** | 4.04* |
| | (2.18) | (2.15) |
| $Contribution_{g(-i),t-1}$ | 0.85*** | 0.97*** |
| | (0.09) | (0.09) |
| $Group\ Size_{g(i),t}$ | 1.38*** | 0.64* |
| | (0.35) | (0.35) |
| $Contribution_{i,t-1}$ | 0.39*** | 0.46*** |
| | (0.08) | (0.08) |
| $Punishment\ Received_{i,t-1}$ | − 0.12 | − 0.01 |
| | (0.10) | (0.10) |
| $IS_{i,t-1}$ | 7.25*** | 8.88*** |
| | (1.57) | (1.65) |
| $Punishment\ Received_{i,t-1} \times IS_{i,t-1}$ | 0.22** | |
| | (0.10) | |
| $Punishment\ Received_{i,t-1} \times \mathrm{IS}_{i,t-1} \times$ | | 0.49 |
| $\mathbb{1}\left(Contribution_{i,t-1} > Contribution_{g(-i),t-1}\right)$ | | (0.33) |
| $Punishment\ Received_{i,t-1} \times \mathrm{IS}_{i,t-1} \times$ | | 0.16* |
| $\mathbb{1}\left(Contribution_{i,t-1} < Contribution_{g(-i),t-1}\right)$ | | (0.09) |
| Observations | 4,800 | 4,800 |
| Log likelihood | -11,558.33 | -11,368.93 |

The table reports “raw” parameter estimates for the panel data (doubly) censored regression models used to compute the marginal effects presented in Table 2. Units of observation are individuals-per-period. All the specifications include subject, period, group, and session random effects. Standard errors are reported in parentheses. Significance levels: *** at 1%, ** at 5%, * at 10%

**Table 12** Parameter estimates fixed-effects censored regression models for contributions

| | (1) | (2) |
|---|---|---|
| $Moving_{i}$ | − 1.74 | − 2.88 |
| | (3.01) | (3.02) |
| $IS_{i,t}$ | − 26.07*** | − 25.19*** |
| | (2.56) | (2.51) |
| $Moving_{i} \times IS_{i,t}$ | − 6.73** | − 5.35** |
| | (2.74) | (2.70) |
| $High\ initial\ group\ contribution_{g(i)}$ | 6.21** | 5.02** |
| | (2.56) | (2.55) |
| $Contribution_{g(-i),t-1}$ | 0.93*** | 1.08*** |
| | (0.09) | (0.09) |
| $Group\ Size_{g(i),t}$ | 2.19*** | 1.70*** |
| | (0.30) | (0.30) |
| $Contribution_{i,t-1}$ | 0.20*** | 0.33*** |
| | (0.08) | (0.08) |
| $Punishment\ Received_{i,t-1}$ | − 0.15 | − 0.05 |
| | (0.11) | (0.10) |
| $IS_{i,t-1}$ | 11.56*** | 15.75*** |
| | (1.98) | (2.04) |
| $Punishment\ Received_{i,t-1} \times IS_{i,t-1}$ | 0.18* | |
| | (0.10) | |
| $Punishment\ Received_{i,t-1} \times \mathrm{IS}_{i,t-1} \times$ | | 0.46 |
| $\mathbb{1}\left(Contribution_{i,t-1} > Contribution_{g(-i),t-1}\right)$ | | (0.33) |
| $Punishment\ Received_{i,t-1} \times \mathrm{IS}_{i,t-1} \times$ | | 0.17* |
| $\mathbb{1}\left(Contribution_{i,t-1} < Contribution_{g(-i),t-1}\right)$ | | (0.10) |
| Observations | 4,800 | 4,800 |

The table reports parameter estimates for fixed-effects censored regression models for contributions using the two-step approach proposed Honoré and Kesina (2017) to identify the coefficients of time-invariant predictors. Units of observation are individuals-per-period. Standard errors are reported in parentheses. Significance levels: *** at 1%, ** at 5%, * at 10%

**Table 13** Parameter estimates Censored regression models for punishment

| | (1) | (2) |
|---|---|---|
| *Intercept* | 14.98*** | 6.93*** |
| | (1.79) | (2.37) |
| $Moving_i$ | 6.24** | 1.73 |
| | (2.53) | (2.60) |
| $Contribution_{i,t}$ | − 1.01*** | − 0.82*** |
| | (0.04) | (0.07) |
| $Contribution_{i,t} \times Moving_i$ | | 0.18** |
| | | (0.08) |
| $max(Contribution_{g(-i),t}$ - $Contribution_{i,t}, 0)$ | | 0.47*** |
| | | (0.07) |
| $max(Contribution_{i,t}$ - $Contribution_{g(-i),t}, 0)$ | | 0.04 |
| | | (0.07) |
| Observations | 2399 | 2399 |
| Log likelihood | – 4573.076 | – 4394.99 |

The table reports "raw" parameter estimates for the panel data (doubly) censored regression models used to compute the marginal effects presented in Table 3. Units of observation are individuals-per-period, with the sample restricted to subjects playing under IS. All the specifications include subject, period, group, and session random effects. Standard errors are reported in parentheses. Significance levels: $***$ at 1%, $**$ at 5%, $*$ at 10%

**Table 14** Parameter estimates Fixed-effects censored regression models for punishment

| | (1) | (2) |
|---|---|---|
| $Moving_i$ | 7.07** | 3.54 |
| | (3.08) | (3.03) |
| $Contribution_{i,t}$ | − 1.12*** | − 0.98*** |
| | (0.05) | (0.09) |
| $Contribution_{i,t} \times Moving_i$ | | 0.23** |
| | | (0.11) |
| $max(Contribution_{g(-i),t}$ - $Contribution\ _{i,t}, 0)$ | | 0.68*** |
| | | (0.09) |
| $max(Contribution_{i,t}$ - $Contribution\ _{g(-i),t}, 0)$ | | 0.09 |
| | | (0.09) |
| Observations | 2,399 | 2,399 |

The table reports parameter estimates for fixed-effects censored regression models for punishment received using the two-step approach proposed Honoré and Kesina (2017) to identify the coefficients of time-invariant predictors. Units of observation are individuals-per-period, with the sample restricted to subjects playing under IS. Standard errors are reported in parentheses. Significance levels: $***$ at 1%, $**$ at 5%, $*$ at 10%

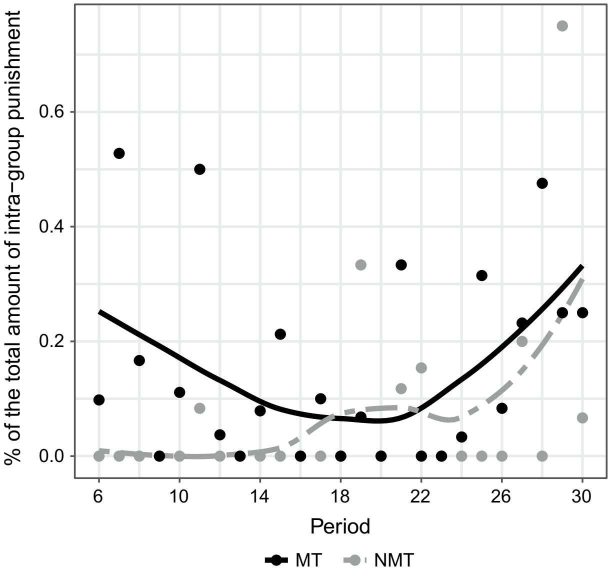



**Fig. 9 Share of anti-social punishment in IS, by period and treatment.** The circles represent the share of anti-social punishment—defined as the proportion of the punishment directed towards subjects contributing more than the group average—per period in groups implementing informal sanctioning institutions under the two treatments. Solid black lines (for MT) and dashed gray lines (for NMT) represent locally weighted scatterplot smoothing curves fitted to the data

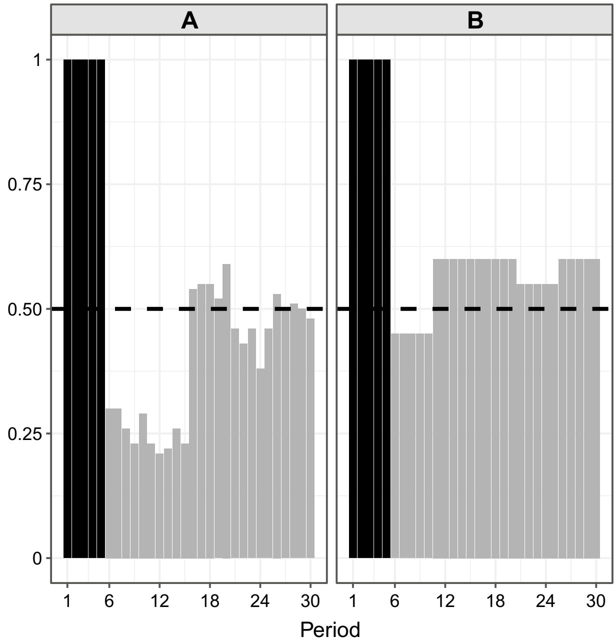



**Fig. 10** **Share of participants in groups implementing IS, by period and treatment**. The bars represent the proportion of subjects located in groups with informal sanctioning institutions in each period under MT (panel A) and NMT (panel B), averaged across sessions. Darker bars correspond to the first 5 periods of the experiment, when groups are fixed and they all implement informal sanctions; lighter bars start in period 6, once subjects are allowed to move between groups and to vote—every 5 periods—on the rule-set to be implemented in their current group (see Sect. 2). The dashed horizontal line corresponds to the situation in which participants are equally distributed between IS and FS

**Table 15** Parameter estimates Panel probit models for migration

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| *Intercept* | − 4.63*** | − 4.57*** | − 4.61*** | − 4.55*** |
| | (0.72) | (0.72) | (0.73) | (0.73) |
| $IS_{i,t}$ | − 0.09 | − 0.08 | − 0.10 | − 0.10 |
| | (0.08) | (0.08) | (0.08) | (0.08) |
| $Contribution_{g(i),t-1} - Contribution_{g(-i),t-1}$ | 1.69*** | 1.68*** | 1.63*** | 1.63*** |
| | (0.39) | (0.39) | (0.39) | (0.39) |
| $Average\ Payoff_{h \neq g,t} - Average\ Payoff_{g,t}$ | 0.94*** | 0.98*** | 0.94*** | 0.98*** |
| | (0.08) | (0.09) | (0.08) | (0.09) |
| $Group\ Size_{g(i),t}$ | − 0.10*** | − 0.10*** | − 0.10*** | − 0.10*** |
| | (0.02) | (0.02) | (0.02) | (0.02) |
| $Vote\ Different\ from\ Group_{i,t-1}$ | 0.27*** | 0.26*** | 0.26*** | 0.26*** |
| | (0.09) | (0.09) | (0.09) | (0.09) |
| $Punishment\ Received_{i,t}$ | 0.38*** | 0.38*** | 0.36*** | 0.36*** |
| | (0.11) | (0.11) | (0.11) | (0.11) |
| $Punishment\ Received_{i,t} \times IS_{i,t}$ | − 0.01 | − 0.01 | − 0.01 | − 0.01 |
| | (0.12) | (0.12) | (0.12) | (0.12) |
| $Different\ Institutions_t$ | | − 0.03 | | − 0.03 |
| | | (0.08) | | (0.08) |
| $Migration_{i,t-1}$ | | | 0.14 | 0.14 |
| | | | (0.09) | (0.09) |
| Observations | 5,000 | 5,000 | 4,800 | 4,800 |
| Log likelihood | –1000.38 | –1000.32 | − 998.99 | − 998.92 |

The table reports "raw" parameter estimates for the panel probit models used to compute the marginal effects presented in Table 4. Units of observation are individuals-per-period. All the models include subject, period, group, and session random effects. Standard errors are reported in parentheses. Significance levels: *** at 1%, ** at 5%, * at 10%

**Table 16** Parameter estimates fixed-effects panel probit models for migration

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| $IS_{i,t}$ | 0.12 | 0.08 | 0.10 | 0.07 |
| | (0.09) | (0.09) | (0.09) | (0.09) |
| $Contribution_{g(i),t-1} - Contribution_{g(-i),t-1}$ | 1.73*** | 1.74*** | 1.70*** | 1.70*** |
| | (0.44) | (0.44) | (0.44) | (0.44) |
| $Average\ Payoff_{h \neq g,t} - Average\ Payoff_{g,t}$ | 0.70*** | 0.70*** | 0.71*** | 0.70*** |
| | (0.09) | (0.09) | (0.09) | (0.09) |
| $Group\ Size_{g(i),t}$ | − 0.14*** | − 0.15*** | − 0.14*** | − 0.14*** |
| | (0.02) | (0.02) | (0.02) | (0.02) |
| $Vote\ Different\ from\ Group_{i,t-1}$ | 0.28*** | 0.28*** | 0.28*** | 0.27*** |
| | (0.09) | (0.09) | (0.09) | (0.09) |
| $Punishment\ Received_{i,t}$ | 0.31** | 0.31** | 0.29** | 0.29** |
| | (0.12) | (0.12) | (0.12) | (0.12) |
| $Punishment\ Received_{i,t} \times IS_{i,t}$ | 0.10 | 0.08 | 0.10 | 0.08 |
| | (0.12) | (0.12) | (0.12) | (0.12) |
| $Different\ Institutions_t$ | | 0.02 | | 0.03 |
| | | (0.09) | | (0.09) |
| $Migration_{i,t-1}$ | | | 0.11 | 0.12 |
| | | | (0.09) | (0.09) |
| Observations | 2325 | 2325 | 2325 | 2325 |
| Log likelihood | − 854.05 | − 853.65 | − 853.84 | − 853.45 |

The table reports parameter estimates from bias-corrected fixed-effects panel probit models for migration using Fernández-Val and Weidner (2016)'s method. The bias correction is obtained from jackknife estimates; using an analytical correction yields virtually identical results. Differences in the number of observations with respect to Tables 4 and 15 are due to the fact that Fernández-Val and Weidner (2016)'s method drops subjects for whom the dependent variable does not change over time. All the models include subject and period fixed-effects. Standard errors are reported in parentheses. Significance levels: *** at 1%, ** at 5%, * at 10%

## References

Andreoni J, Gee LK (2012) Gun for hire: delegated enforcement and peer punishment in public goods provision. J Public Econ 96(11):1036–1046
Cobo-Reyes R, Katz G, Meraglia S (2019) Endogenous sanctioning institutions and migration patterns: experimental evidence. J Econ Behav Org 158:575–606
Dal Bo P, Foster A, Putterman L (2010) Institutions and behavior: experimental evidence on the effects of democracy. Am Econ Rev 100(5):2205–29
Dannenberg A, Gallier C (2020) The choice of institutions to solve cooperation problems: a survey of experimental research. Exp Econ 23:716–49
De Stefano V (2016) The rise of the just-in-time workforce: on-demand work, crowdwork, and labor protection in the gig-economy. Compar Labor Law Policy 37(1):471–504
Dehingkar P, Anderson E (2004) People on the move. New policy challenges for increasngly mobile populations. Nat Resour Perspect, 92. London: Overseas Development Institute
Dessi R, Piccolo S (2016) Merchant guilds, taxation and social capital. Eur Econ Rev 83:90–110
Ertan A, Page T, Putterman L (2009) Who to Punish? Individual decisions and majority rule in mitigating the free rider problem. Eur Econ Rev 53(5):495–511
Fehr E, Schmidt KM (1999) A theory of fairness, competition and cooperation. Quart J Econ 114(3):817–868
Fehr E, Gächter S (2000) Cooperation and punishment in public goods experiments. Am Econ Rev 90(4):980–994
Fernández-Val I, Weidner M (2016) Individual and time effects in nonlinear panel models with large N. T. J Econ 192(1):291–312
Fréchette G (2012) Session-effects in the laboratory. Exp Econ 15(3):485–498
Greene W (2003) Econometric analysis. Prentice Hall, Upper Saddle River, NJ
Gürerk O, Irlenbusch B, Rockenbach B (2006) The competitive advantage of sanctioning institutions. Science 312(5770):108–111
Gürerk O, Irlenbusch B, Rockenbach B (2014) On cooperation in open communities. J Public Econ 120(1):220–230
Harris JR, Todaro M (1970) Migration, unemployment and development. Am Econ Rev 60(1):126–142
Herrmann B, Thöni C, Gächter S (2008) Antisocial punishment across societies. Science 319(5868):1362–1367
Hirschman A (1970) Exit, voice and loyalty. Responses to decline in firms, organizations and states. Harvard University Press, Cambridge, MA
Honoré B, Kesina M (2017) Estimation of some nonlinear panel data models with both time-varying and time-invariant explanatory variables. J Bus Econ Stat 35(4):543–558
Kamei K, Putterman L, Tyran JR (2015) State or nature? Endogenous formal versus informal sanctions in the voluntary provision of public goods. Exp Econ 18(1):38–65
Kripfganz S, Schwarz C (2019) Estimation of linear dynamic panel data models with time-invariant regressors. J Appl Economet 34(4):526–46
Markussen T, Putterman L, Tyran JR (2014) Self-organization for collective action: an experimental study of voting on sanction regimes. Rev Econ Stud 81(1):301–324
Markussen T, Putterman L, Wang L (2017) Governing collective action in the face of observational error. Brown Economics Working Paper Series
Nicklisch A, Grechenig K, Thöni C (2016) Information-sensitive leviathans. J Public Econ 144(1):1–13
Ogilvie S (2011) Institutions and European trade. Merchant guilds, 1000–1800. Cambridge University Press, Cambridge
Ostrom E, Walker J, Gardner R (1992) Covenants with and without a Sword: self-governance is possible. Am Polit Sci Rev 86(2):404–417
Page T, Putterman L, Unel B (2005) Voluntary association in public goods experiments: reciprocity. Mimicry Efficiency Econ J 115(506):1032–1053
Poen E (2009) The tobit model with feedback and random effects: a Monte-Carlo study. CeDEx Discussion Paper Series 2009-14, University of Nottingham
Robbett A (2014) Local institutions and the dynamics of community sorting. Am Econ J Microecon 6(3):136–156
Salmon T, Weber R (2017) Maintaining efficiency while integrating entrants from lower performing groups: an experimental study. Econ J 127:417–444

Smith A (2013) Estimating the causal effect of beliefs on contributions in repeated public goods. Exp Econ 16(3):414–425

Sutter M, Haigner S, Kocher MG (2010) Choosing the carrot or the stick? Endogenous institutional choice in social dilemma situations. Rev Econ Stud 77(4):1540–1566

Tiebout CM (1956) A pure theory of local expenditures. J Polit Econ 64(5):416–424

Tyran J-R, Feld LP (2006) Achieving compliance when legal sanctions are non-deterrent. Scand J Econ 108(1):135–156

Weber R (2006) Managing growth to achieve efficient coordination in large groups. Am Econ Rev 96(1):114–126

Wooldridge J (2005) Simple solutions to the initial conditions problem in dynamic, nonlinear panel data models with unobserved heterogeneity. J Appl Economet 20(1):39–54

Wooldridge J (2010) Econometric analysis of cross section and panel data. MIT Press, Cambridge, MA

Zhang B, Li C, De Silva H, Bednarik P, Sigmund K (2014) The evolution of sanctioning institutions: an experimental approach to the social contract. Exp Econ 17(2):285–303